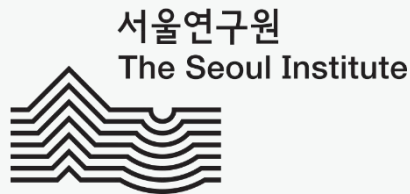


Developing Transport Infrastructure in Seoul

Chang Yi

Research Fellow



Seoul ?



9.3mil.
(Capital region 25mil)



3.1mil.
(Capital region 4.7mil)



327.1km
(9 lines)



9,334
(629 route)



72,109



Seoul in the past (after the Korean War)



Setting the Stage for Seoul: the Rapid Growth after the Korean War



Setting the Stage for Seoul: the Rapid Growth after the Korean War

Urban planning area increased almost twice to 605km² from 135.4km² (1913-1973)



Year	Population	Household	Housing (unit)	Housing Shortage (percent)	Illegal Units	GNP/capita (US\$)
1926	306,363	68,682	64,889	5.8		
1935	636,995	131,239	101,767	22.5		
1939	930,547	154,223	-	-		
1944	1,078,178	220,938	132,000	40.3		
1950	1,693,224	318,673	-	-	-	67
1960	2,445,402	446,874	275,436		40,000	79
1970	5,433,198	1,096,871	600,367		200,000	253
1980	8,364,379	1,849,324	968,133		-	1,597
1990	10,612,577	2,820,292	1,430,981		94,974	5,833
1995	10,595,943	3,448,466	1,863,466		73,500	10,037

Right after the Korean War, there was no concept of “urban planning”.

Building Roadways like Crazy – the Legacy of Kim Hyun-ok (1966-1970)

Mayor: Mr. Kim Hyun-ok (1966-1970)

- A former military man nicknamed “Bulldozer”
- A field commander committed to the “battle of urbanization.”
- A well-qualified person who could resolve the transportation and other urban problems



1961

1966

5-year national economic development plan

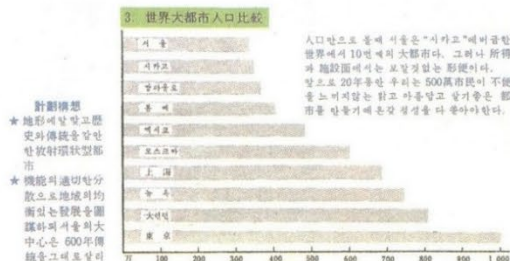
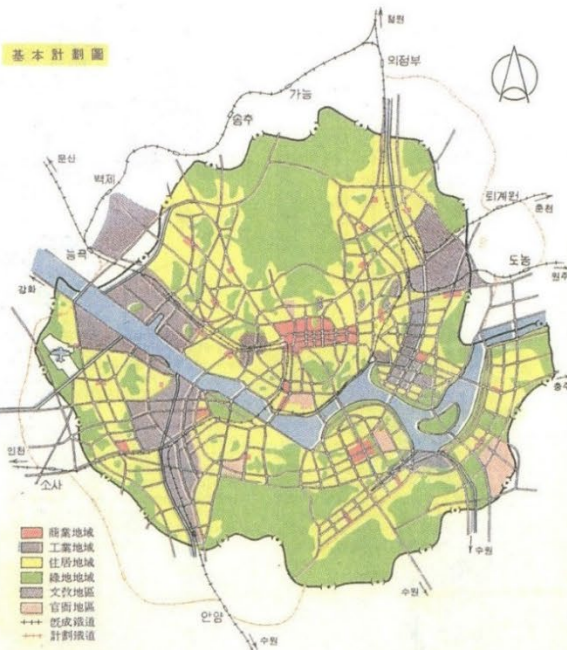
The expansion and growth of Seoul

Population
Increasing by
300,000 for 3 years

Various infrastructure development
projects to solve the long-standing
problems

To lead the
modernization of
the country

The First Urban Master Plan for Seoul

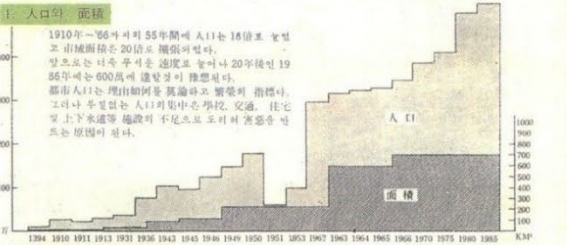
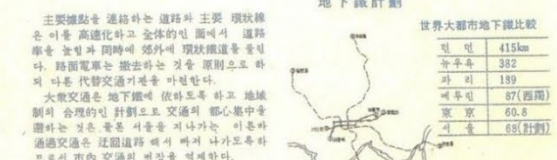
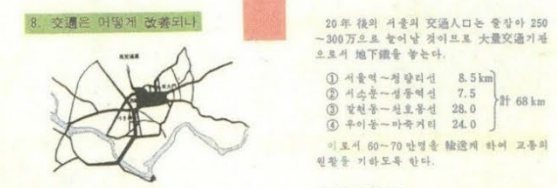
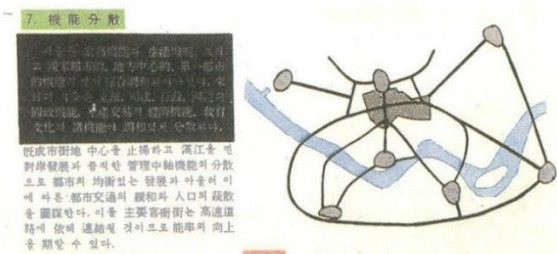


4. 人口分散은 어떻게 한다

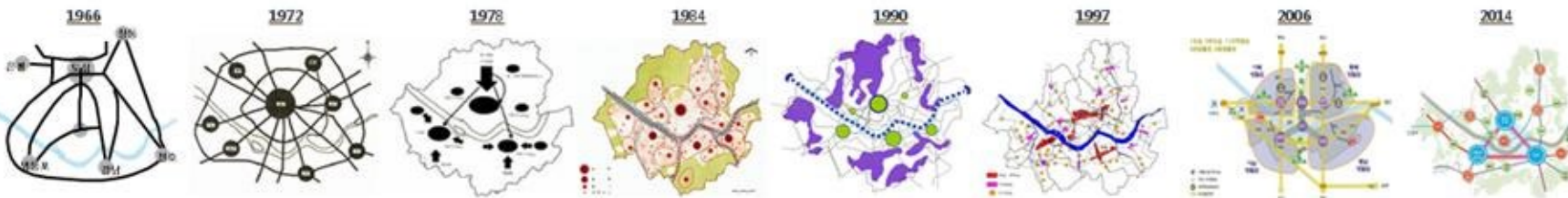
도시가 팽창하면 도시가 팽창하면 사람이 몰려들기 마련이다. 그러나 그렇다고 무제한한 인구의 집중은 여러가지 면에서 불이익을 가져오게 된다. 주택, 학교,水道, 市場, 交通에 지르기까지 도시의供給이 뒤따르지 못하여는一大混亂을 동반하게되므로 도시는 그 정도를 적당한 收容能力이 있으므로 이점을 넘어서는 않된다. 이러한觀點에 立脚해서 이번 計劃에서는 적체로 인구를 끌어들리는 가장 중요한 原因이 되고있는 官廳, 會社等 이른바 管理中軸機能을 다른 適當한 곳으로 옮기는것이다. 一隅을둘런 外에 서울에 있을必要가없는 官廳, 學校, 刑務所 또는 專賣稅工場等을 移轉하는것이다.

5. 人口配分計劃

區分	基準年度人口(1960)	面積	計 劃	備 考
中心市街地	2,800,000	250km ²	2,000,000	150,000,000
郊外地域	200,000	500	2,300,000	70,2,100
綠地帶	400	1,300	600	10,200
1차의 주변地域	1,600	4,100	2,000	15,400
2차의 주변地域	3,000	16,400	3,100	6,100
計	8,000	22,550	10,000	2,000

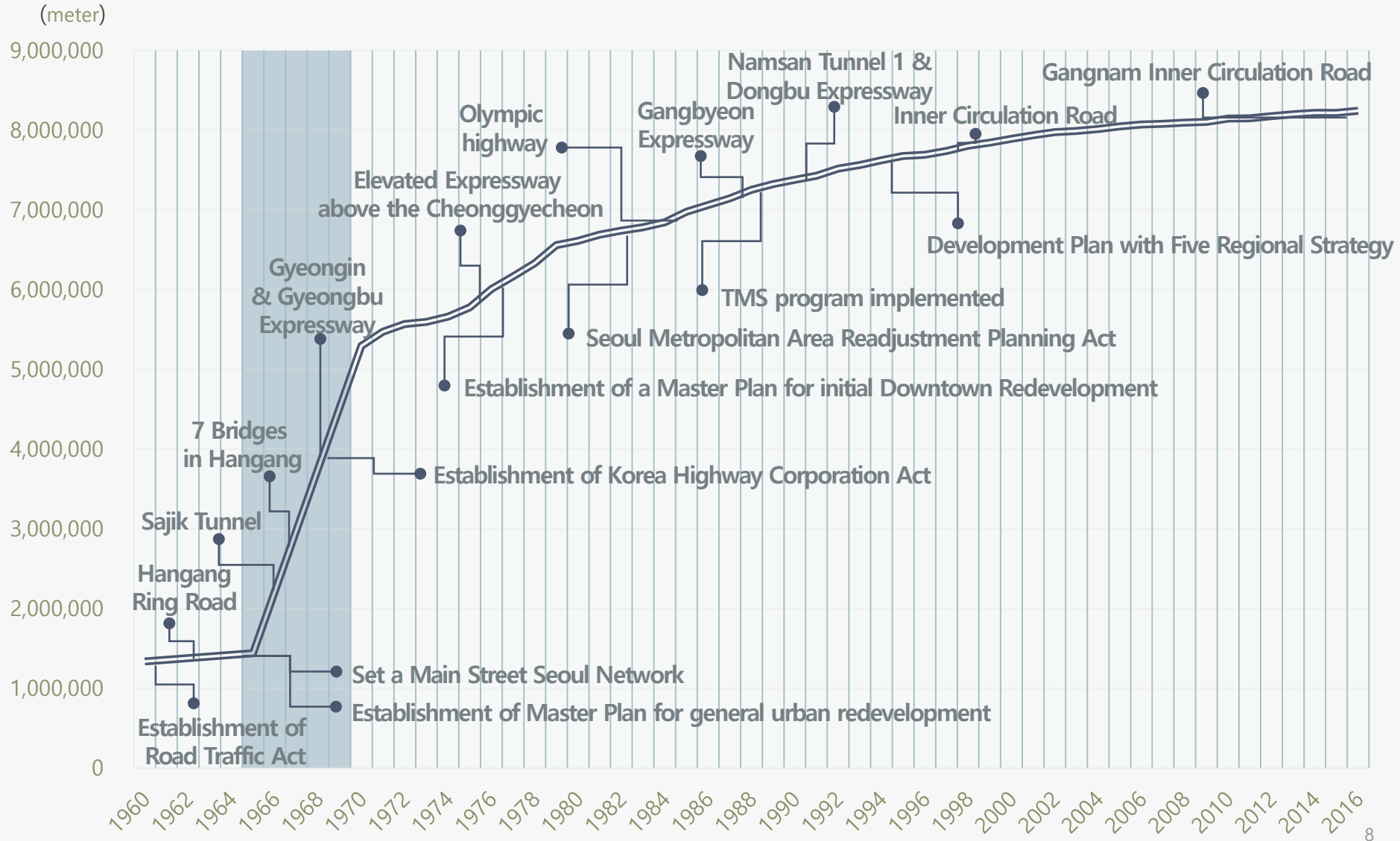


In the 1960s, the government first introduced an urban planning system and established the first urban master plan on its own.

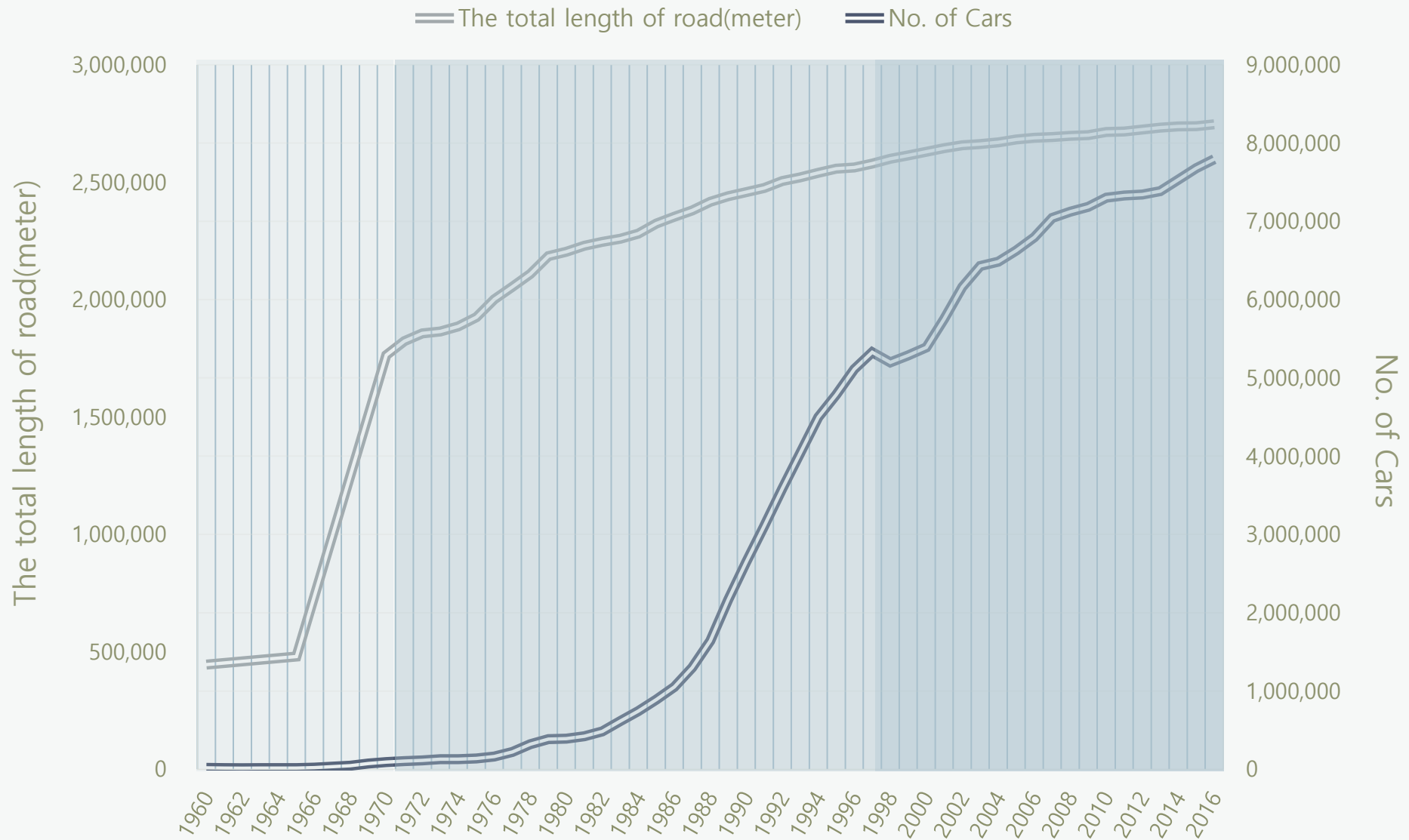


A Significant Increase in the Total Length of Roads

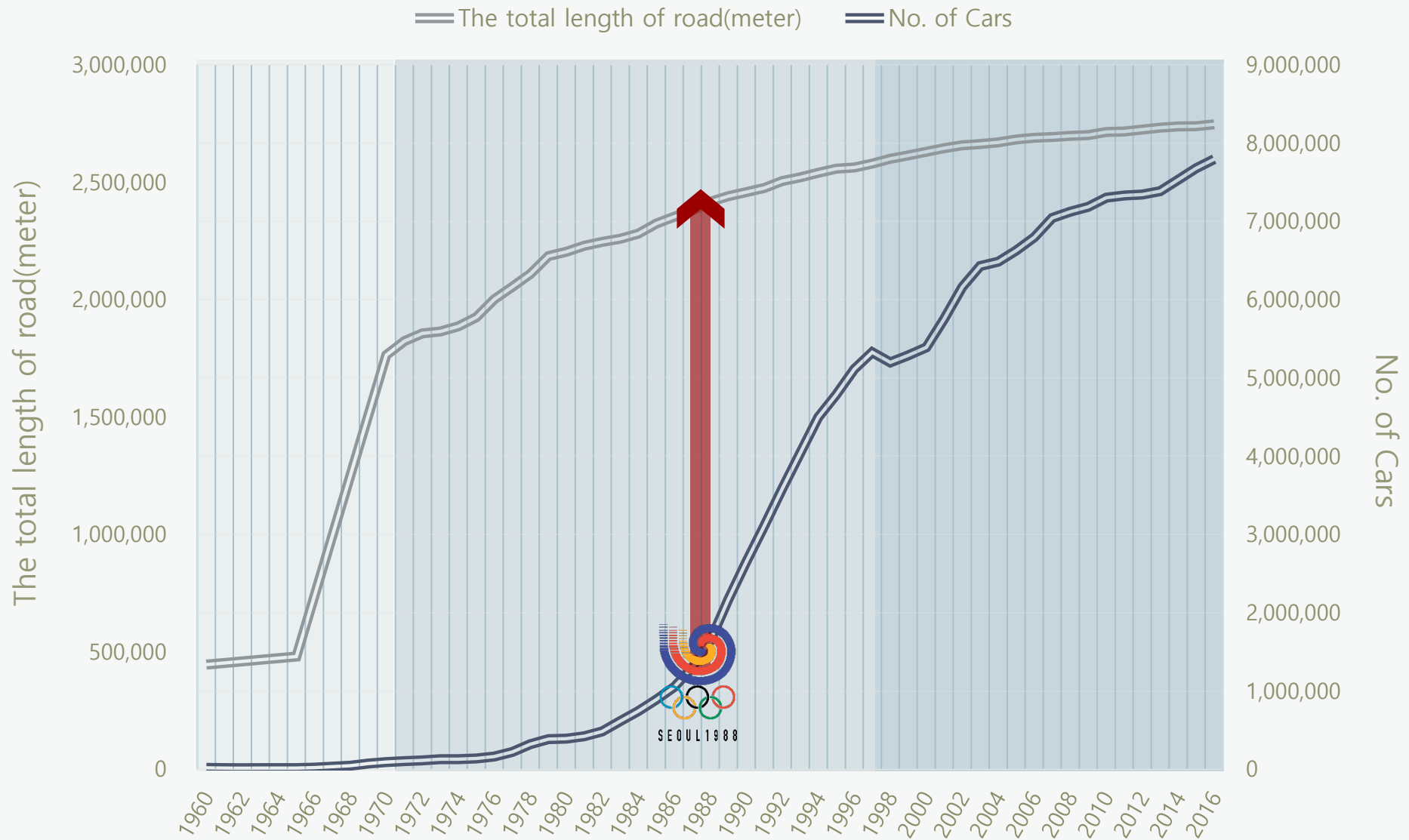
THE TOTAL LENGTH OF ROAD



The Total Length of Road Compared with no. of Cars



The Total Length of Road Compared with no. of Cars



Preparing for the Seoul Olympics

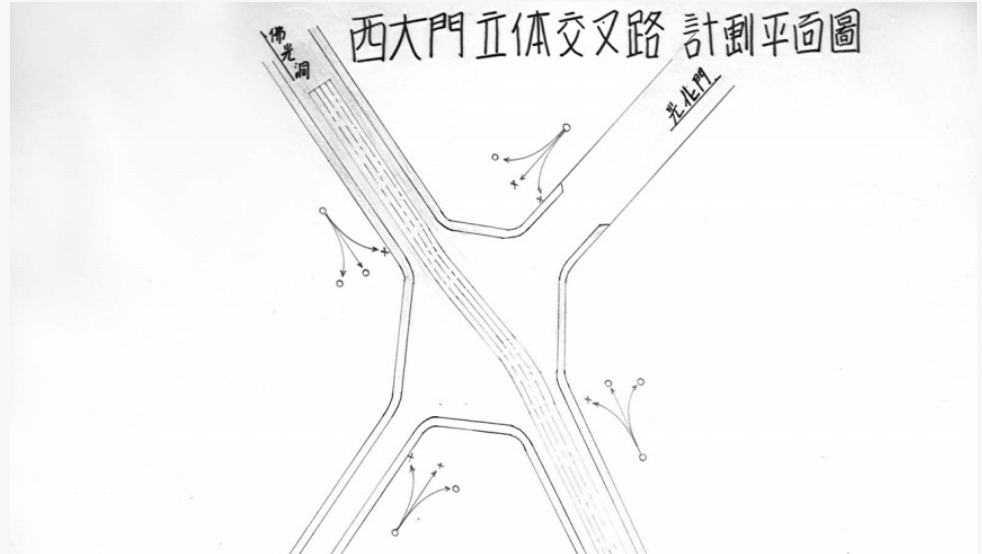


- The **urban highway network** in Seoul had its basic frame completed during the preparation for the Olympics.
- Mayor Goh Kun (1988-1990) suggested building an **inner-ring road bypassing the CBD**
- Planning the **outer beltway** was relatively simple because the areas where the road would be built were greenbelt areas

Transportation Policies Diversified in the 1990s : Quantity → Quality



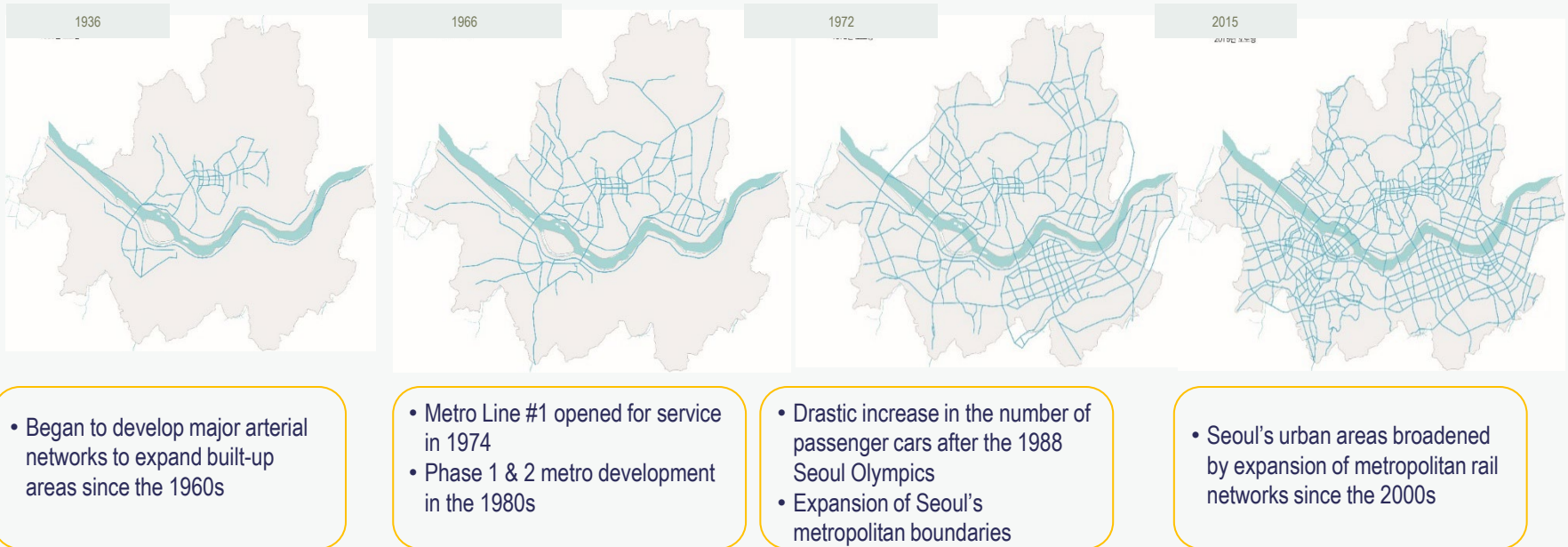
- ◀ A ten-year comprehensive parking management plan
- ▼ To improve the traffic capacity of intersections



- Express city buses were increased and most city buses were equipped with air conditioning
- Transportation System Management (TSM)
- A number of new bridges were constructed
- Traffic demand management
 - ✓ A major contributors to transportation problems had to take responsibility for the expenses
- To improve transport safety
 - ✓ School zones, The driver's license system, Children's Transportation Park

Major urban developments that substantially transformed Seoul's urban structure

Road network development



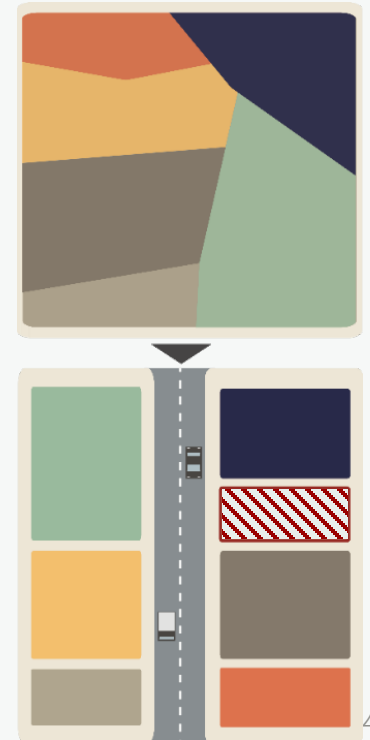
Funding Infrastructure Projects from New Developments

Where did the money come from to build all these roadways?

In the early 1960s, the SMG attempted to reorganize the urban areas through **the land parcel readjustment program**



1. To secure a budget for the plan, the SMG was handed over the rights of properties from the original owners of the target land
2. The SMG **rezoned the land parcels** suitable for urban development
 - Then the land was redistributed to the owners, but the size of redistributed land parcels was smaller than the original land
 - However, there was no backlash from the original owners because they could expect **a sharp increase in the property value thanks to new roadways**



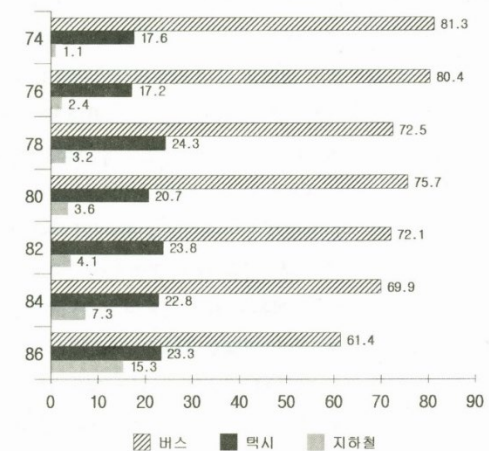
Developing a Subway Transit

A subway system started its operation in 1974

Shovel-ready for building the Subway Line 1

- ✓ In urgent need of a new transit system that enables mass movement of people
- ✓ The city's first subway line was completed in three years and four months, recording **the world's shortest construction period at the time**

The modal share of taxis, buses, subway



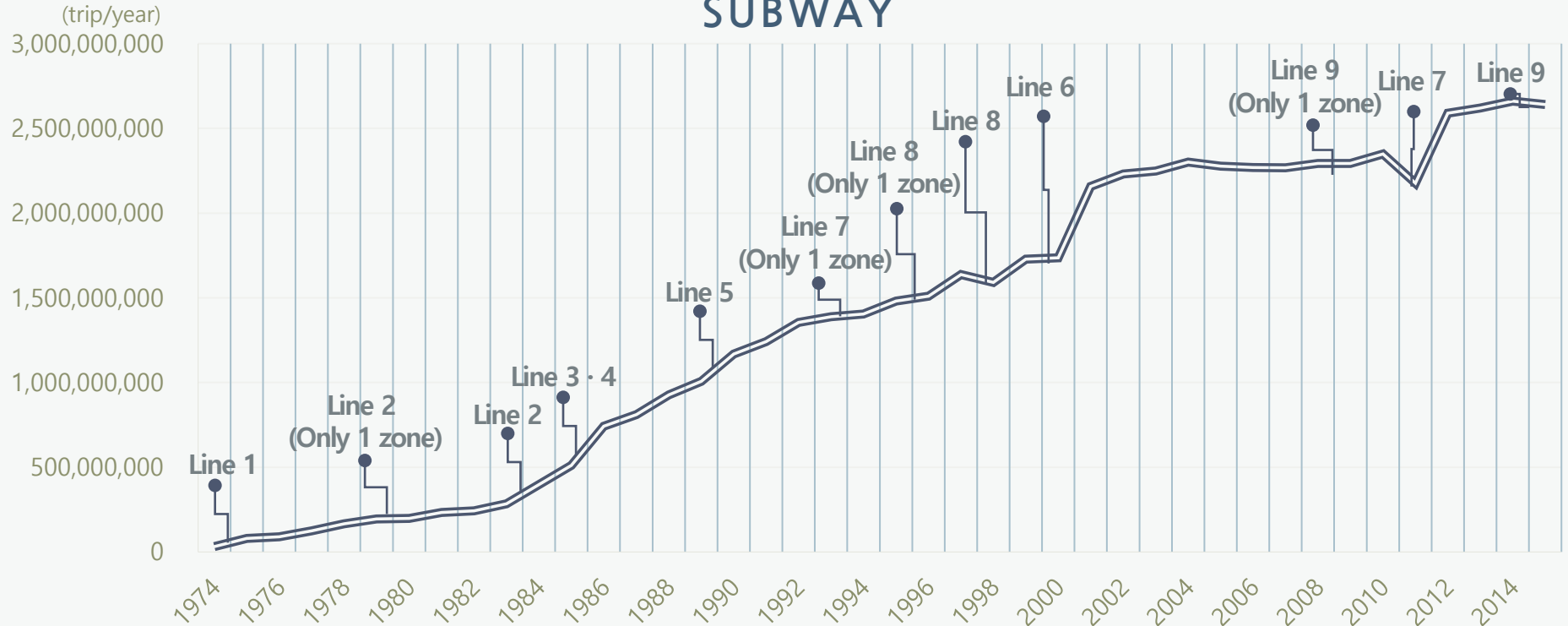
The Circular Subway Line 2 and its Impact

- ✓ In 1974, mayor of Seoul decided to build **a circular rail network** to realize the vision of three nucleus city
- ✓ After the opening of Line 2, the modal share of subway continued to increase, recording 7.3% in 1984, and soaring to 15.3% in 1986 with Line 2, 3, and 4 in full operation
- ✓ The subways were able to carry 2 million passengers to the city center within 40 minutes



Developing a Subway Transit

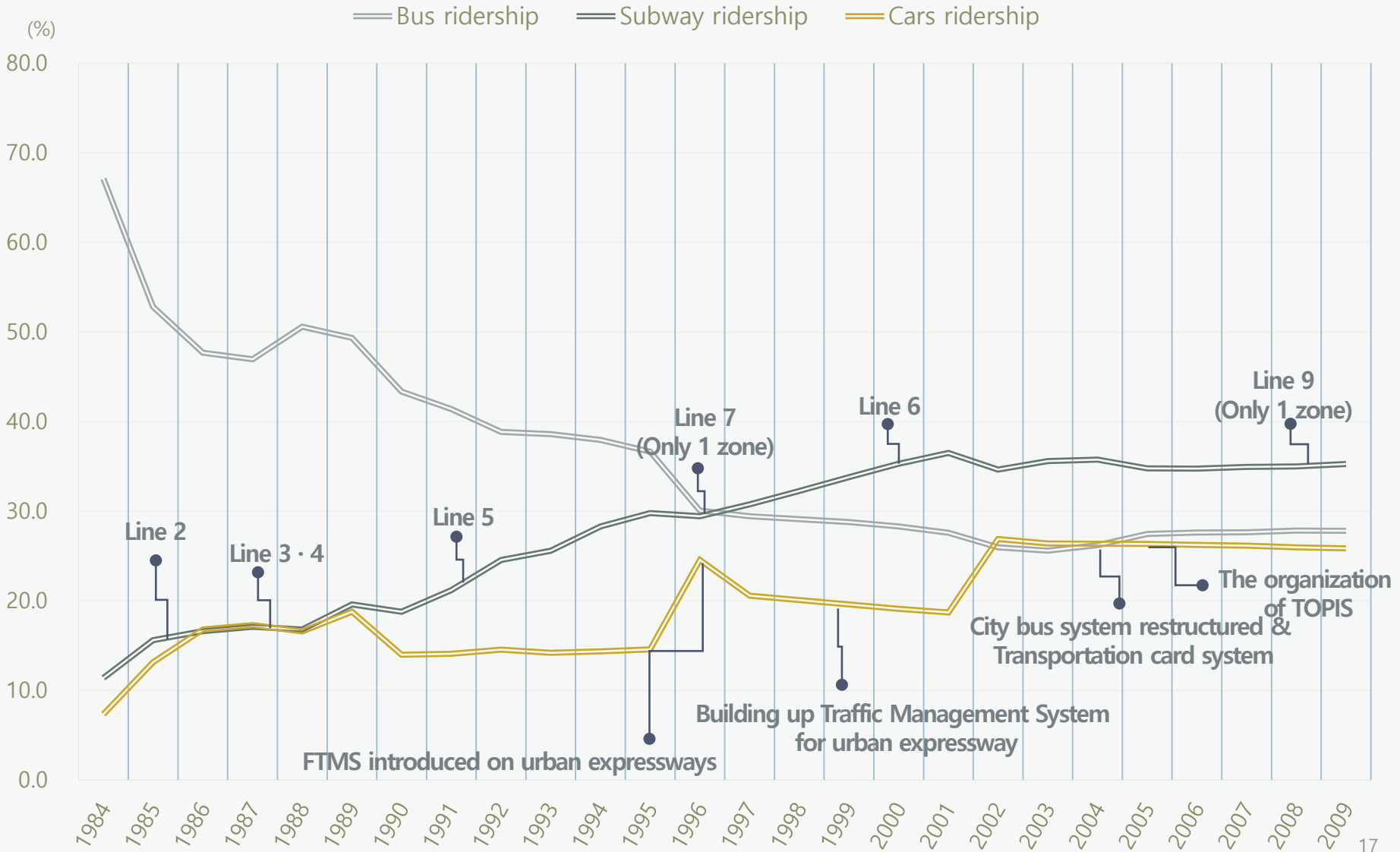
SUBWAY



	Section	Length	the date of the start of construction work	The date of opening	No. of subway stations
Line1	Seoul Station-Cheongnyangni Station	7.8	1971-04-12	1974-08-15	10
Line2	Seongsu Station-Seongsu Station (circuration), (Sindorim Station-Kkachisan Station (branch line), Sinseol-dong Station-Seongsu Station (branch line))	60.2	1978-03-09	1984-05-22	50
Line3	Jichuk Station-Ogeum Station	38.2	1980-02-29	1985-10-18	34
Line4	Danggogae Station -Namtaeryeong Station	31.7	1980-02-29	1985-10-18	26
Line5	Sangil-dong Station, Macheon Station-Banghwa Station	52.3	1990-06-27	1996-03-20	51
Line6	Yeokchon Station-Sinnae Station	35.1	1994-01-08	2000-12-15	38
Line7	Jangam Station-Onsu Station	46.9	1990-12-28	1996-10-11	42
Line8	Amsa Station-Moran Station	17.7	1990-12-29	1996-11-23	17
Line9	Gaehwa Station-Sinnonhyeon Station	26.9	2001-12-31	2009-07-24	25

Developing a Subway Transit

MODE SHARE



Subway - Construction Timeline



Era of Tram

Planning & Introduction

Expansion

Improvement

Before 1970's

1970's

1980's

1990's

2000's

1899 Tram Opened

1968 Tram Ended

1970 Founding Subway
Construction Headquarter

1971 Starting Construction of Line 1
Line 1 opened

1974 Starting Construction of

1978 Line 2

1980 Starting Construction of Line 3,4
Starting service of Line 2

1980 Founding Seoul Metro Co.

1981 Completing Line 2 construction

1984 Line 3,4 opened

Subway Construction Headquarter

1985

1985

1990 Starting Construction
of Line 5

1990 Starting Construction
of Line 7,8

1994 Starting Construction
of Line 6

1994 Founding SMRT Co.

1995 Line 5 Opened

1996 Line 7 Opened

1996 Line 8 Opened

2000 Line 6 Opened

2002 Starting Construction
of Line 9

2009 Line 9 opened

2009 Starting Construction of
UI-Sinseol LRT

Subway - Planning Stage

Early Plan

Y1962 Planning Line 1

The First 5-Year
Economic Development
Plan
(complement)

Y1965 Planning 4 Lines

10-Year Plan for City
Administration

Y1966 Planning 4 Lines

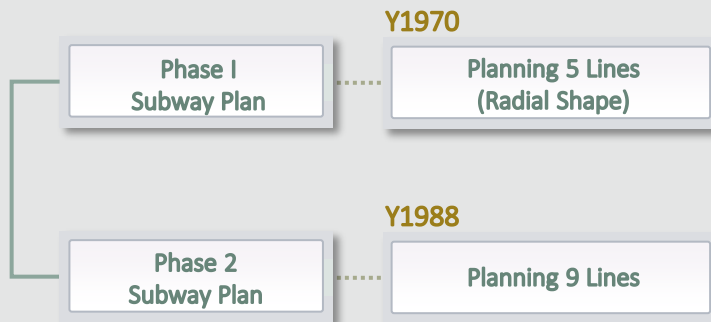
The City
Master
Plan

Y1970 Planning 5 Lines
Initiating
Construction of
Line 1

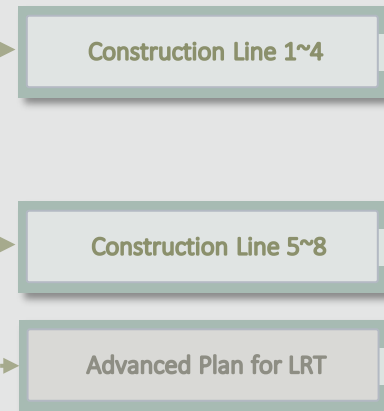
Metropolitan Transport
Plan

Implementation

Plan



Implementation

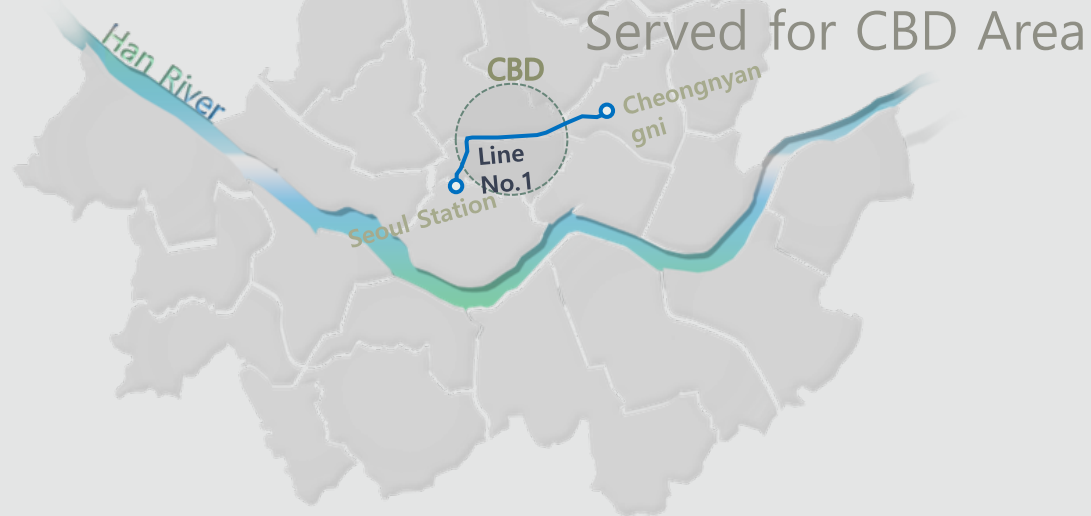


Subway - Introduction Stage

Subway for CBD

Subway Lines

- **1971** • Starting Construction of Line 1
- **1974** • Service Open on Line 1 (Seoul Station ~ Cheongnyangni, 7.8 km)

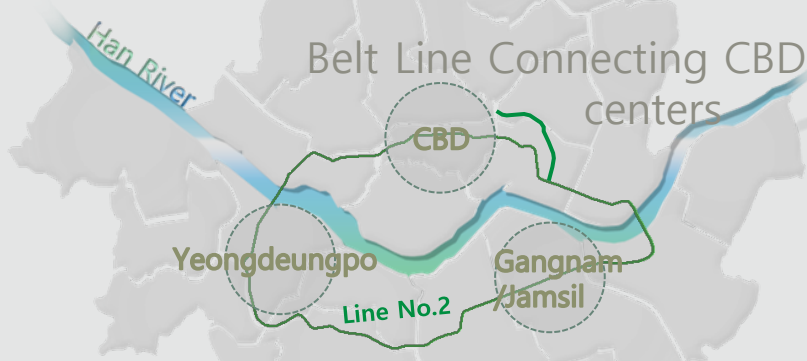


Remark

- Introducing Electrification System to Existing Railways (Gyeong-in Line, Gyeong-won Line, Gyeong-bu Line) and Integrated Operation with Line 1
- Construction completed as scheduled (less than 4 years)

Subway - Expansion Stage

Connecting Sub-centers

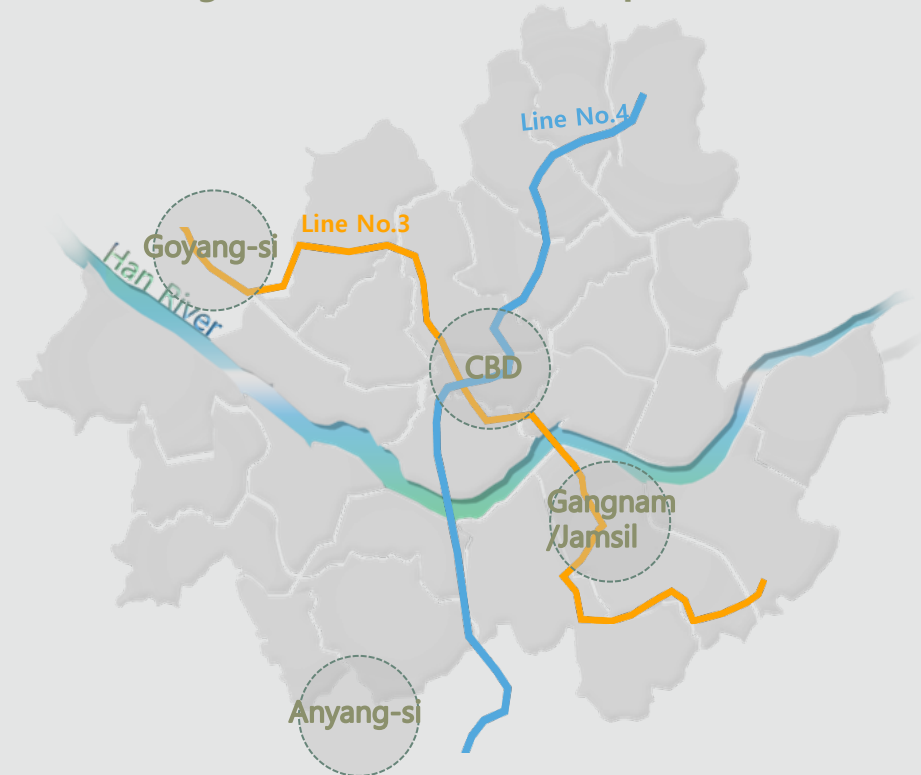
Subway Lines	● 1978 • Starting Construction of Line 2 ● 1980 ~ • Starting service on Line 2 (Opened all sections in 1984)  Belt Line Connecting CBD and Sub-centers Seoul 3 Cores • Inner circular Line in Early plan → Plan changed for Big Belt Line Connecting 3 City centers (48.8 km)
Remark	• Phased Open (Y1980~1984) 

Subway - Expansion Stage

Passing through Radial Route

Subway
Lines

- 1980 • Starting Construction of Line 3, 4
- 1985 ~ • Starting service on Line 3,4 (Opened all sections in 1984)



3,4 Line Connecting CBD and Satellite passing through Radial route

Subway - Expansion Stage

Networking

Subway Lines

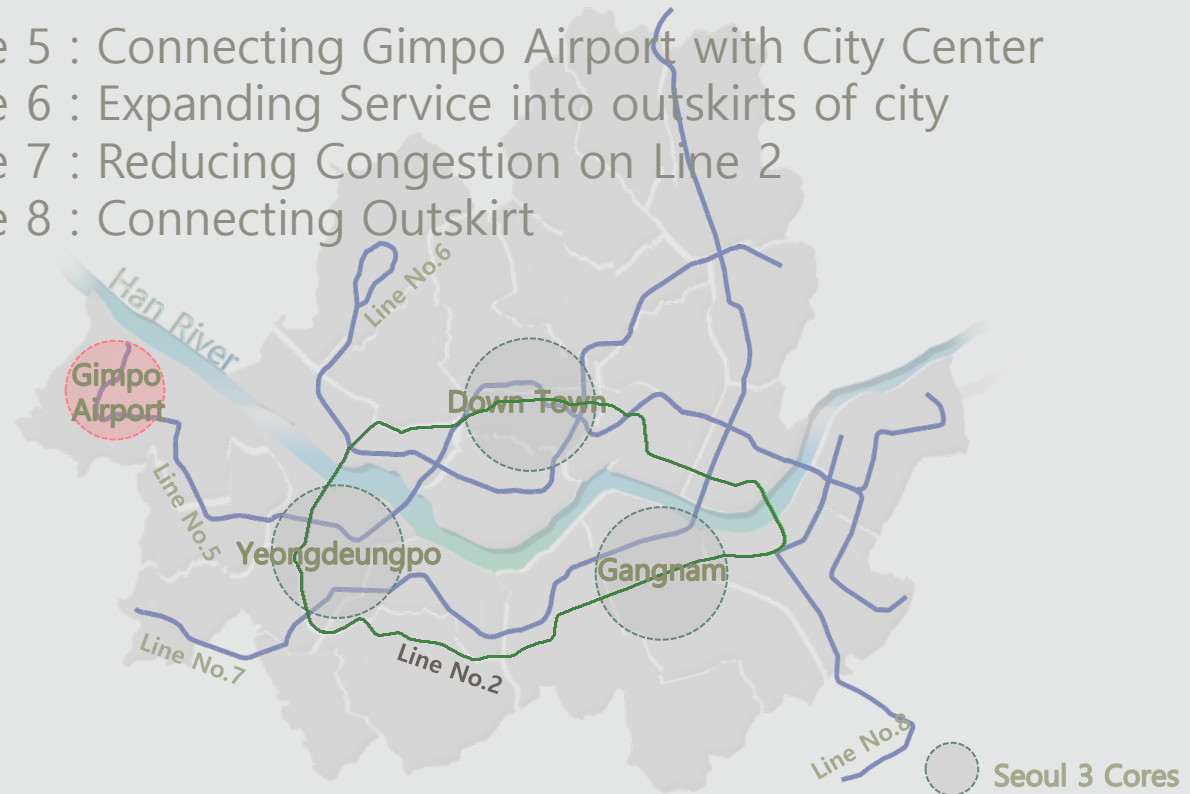
- Y1990 ~ Y1994 • Starting Construction of Phase 2 lines (Line 5~8)
- Y1995 ~ Y2000 • Completed Phase 2 Construction (162.2 km)

Line 5 : Connecting Gimpo Airport with City Center

Line 6 : Expanding Service into outskirts of city

Line 7 : Reducing Congestion on Line 2

Line 8 : Connecting Outskirt



Subway Improvements

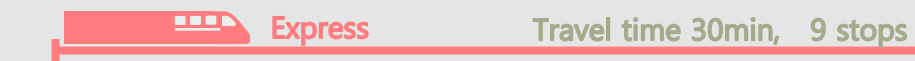
Subway Lines

- **Y2002** • Starting Construction of Line 9
- **Y2009** • Service Started on Line 9 (Gaehe ~ Sinnonhyeon, 27 km)



Remark

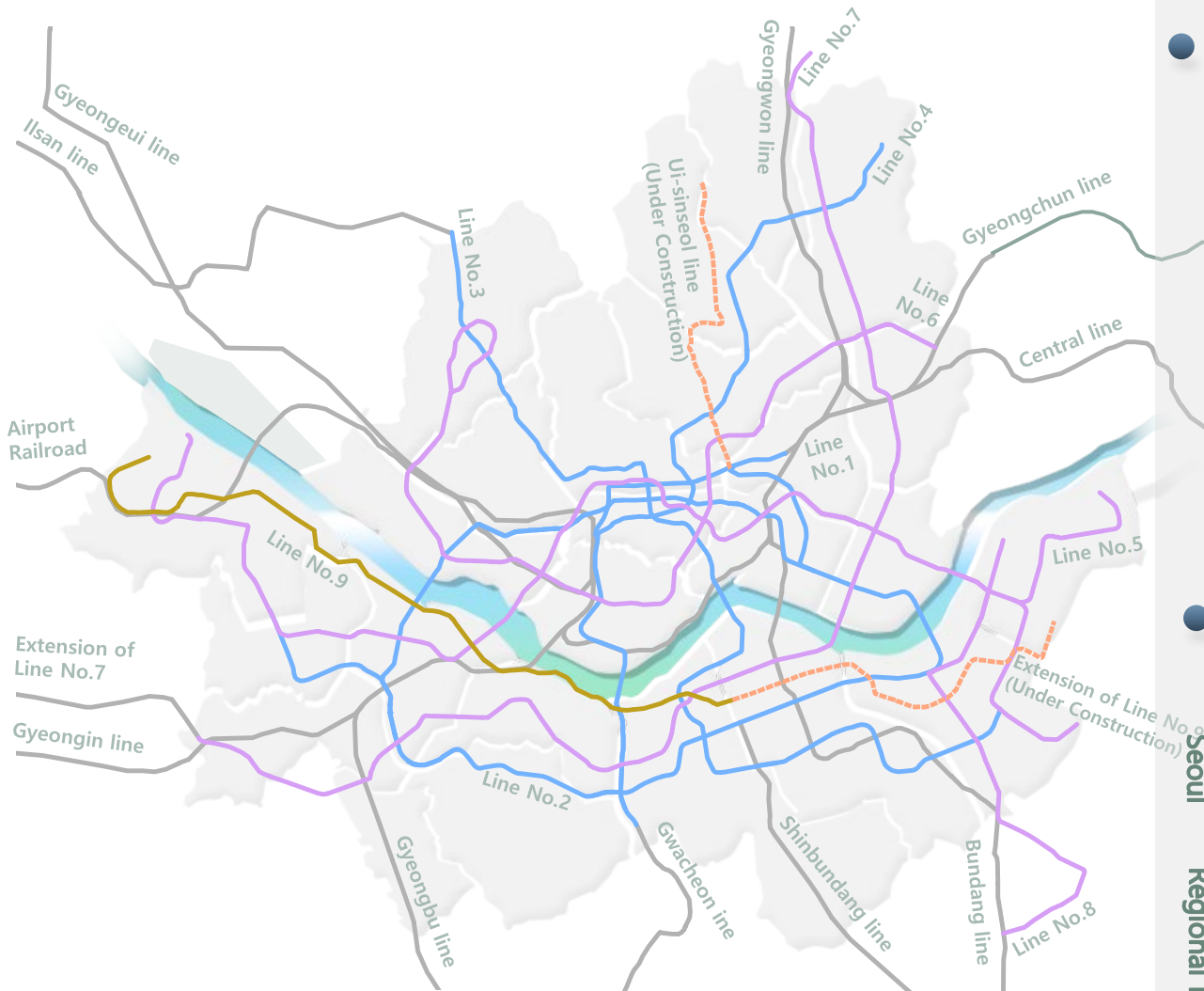
- Skip & Stop (Express and Regular Train Service)
- 24 Minutes faster with Express



■ Stops

○ Express Stops

Current Subway Network



Service Length

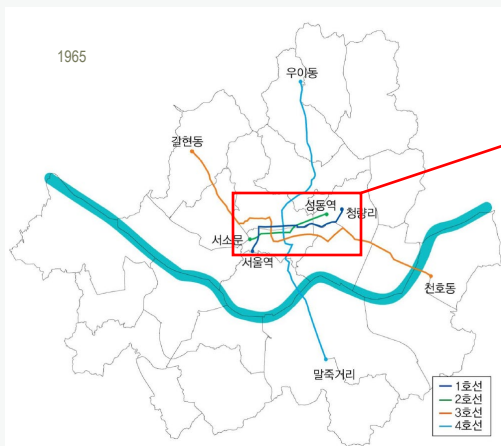
- Line 1~4 137.9 km
- Line 5~8 162.2 km
- Line 9 27.0 km
- Regional /Commuter Rail (12 Lines in Seoul) 327.1 km
- Total Length (Line 1~9) 327.1 km
- Being Constructed 25.0 km
(Extension of Line 9, Ui-dong~Singseol-dong Line)

Operator

Seoul	Seoul Metro	Line 1~4
	SMRT	Line 5~8
	Metro 9 (PPP)	Line 9
Regional Rail	Korail	Regional Rail
	Neo Trans Corp.	Sinbundang Line

Major urban developments that substantially transformed Seoul's urban structure

Metro development



Railroad authority's plan for Metro Line 1 in 1962 (Draft)

- Apr. 1964: Necessity for metro development proposed by the National Assembly Subcommittee on Transport and Communications
- 1965: Plan to build 4 metro lines announced as part of the 'ten-year plan for the city'
 - Line 1: Seoul Station-Cheongnyangni connected (pursuant to the railroad authority's draft metro plan of 1962)
 - Line 2: Seosomun-Seongdong Station connected
 - Line 3: Eunpyeong-Cheonho subcenters connected
 - Line 4: Sungin - Gangnam subcenters connected
- 1966: Incorporated in the Seoul Urban Master Plan

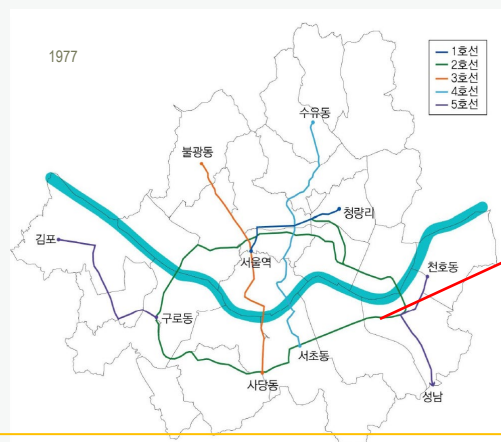


- Metro development plan made more concrete since April 1970
- Mar. 1971: Metro Line 1 construction plan finalized & announced
- Dec. 1972: Line 2 - 5 routes finalized: Most routes were changed, expanding into the areas south of Han River and connecting the old city center within the old city walls and subcenters
 - Line 2: Yeongdeungpo - Wangsimni ← Expanded to Yeongdeungpo
 - Line 3: Mia - Toegyero - Bulgwang-dong ← Changed into a U shape
 - Line 4: Poi-dong, Gangnam - Yulgokno - Daerim-dong ← Into an n shape
 - Line 5: Yeonhui-dong - Cheonho

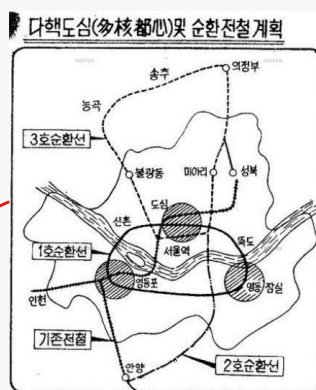
- Seoul Metro Line 1 opened for service on March 15, 1974 (Seoul Station - Cheongnyangni)

Major urban developments that substantially transformed Seoul's urban structure

Metro development



- Metro routes announced in 1972 fully revised
- Circular metro routes that connect three nuclei to build 3 nuclei in Seoul
- Three circulate routes announced in March 1975
- Construction plan for Metro 2 – 5 announced in October 1977
 - Line 2: Circular plan finalized, connecting all 3 nuclei
 - Line 3, 4: Pass through CBD, transit to Line 2, connection of Gangnam & Gangbuk
 - Line 5: Seoul's eastern/western outskirts with Line 2: East (Cheonho – Jamsil - Seongnam), West (Guro – Gimpo)
- Pass through most CBDs and subcenters in the urban master plan of 1978, aligned with multicentric CBDs



Circular metro connecting 3 nuclei (1975)

- Line 2 opened for service in phase from 1980 to 1984
- Oct. 1980: Seoul's east (Sinseong-dong – Sports Complex)
- Dec. 1982: Seoul's southeast (Sports Complex – Seoul Nat'l Univ. of Education)
- Sep. 1983: CBD → northeast (Euljiro-1-ga – Seongsu-dong)
- Dec. 1983: Seoul's southeast → south (Seoul Nat'l Univ. of Education – Seoul Nat'l Univ.)
- May 1984: Seoul's south → CBD (Seoul Nat'l Univ. – City Hall)

Promoting Gangnam / Jamsil development & population decentralization in CBD

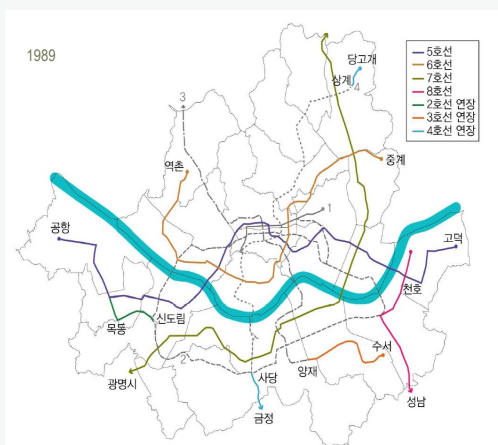


- Feb. 1977: Line 3, 4 routes changed and finalized
- Crossing the CBD in an X pattern to adapt to the changing metropolitan environment
- Transfer on Line 2 in Gangbuk / Gangnam → Acceleration of 3-nuclei development
- 1978: Pass through some centers and subcenters in the Seoul Urban Master Plan pursuant to the multicentric structure plan
- Line 3: Seoul's northwest → southeast (Byeokje – Gupabal – Chungmuro – Yangjae)
- Line 4: Seoul's northeast → south (Sanggye – Dongdaemun – Seoul Station – Sadang-dong)
- Commenced simultaneously on February 29, 1980
- 1981: Some Line 3, 4 routes changed

- October 18, 1985: Line 3, 4 fully open for service

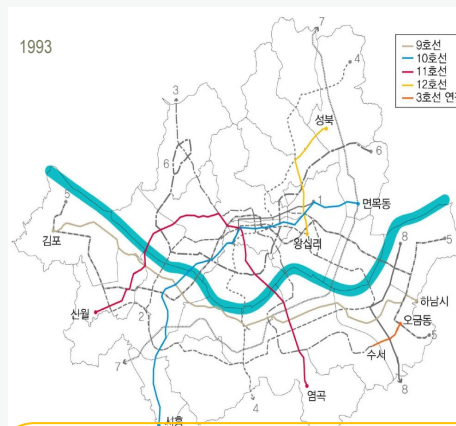
Major urban developments that substantially transformed Seoul's urban structure

Metro development



- Road traffic & growing metro users led to severe congestion since the late 1980s
- 1988: Additional metro construction plan developed & commissioned to the Korea Research Institute for Human Settlements (KRIHS)
- 1989: Phase 2 metro plan announced: Plan for Line 5 – 8, extension of Line 2 – 4, more metro lines that do not pass through CBD (Line 5 excluded)
 - Line 5: Yeongdeungpo, Yeouido, CBD connected; first to pass through Yeouido
 - Line 6: More transfer stations (to existing metro lines) → Enhanced metro connectivity and accessibility
 - Line 7: Seoul's northeast → southwest, Gangnam & Yeongdeungpo connected
 - Line 8, Line 2 – 4 extension: Connected to major new cities and land development sites

- Nov. 1995: Line 5 opened for service
- 1996: Line 7, 8 opened for service
- 2000: Line 6 opened for service



- Jan. 1991: Metro Phase 3 first discussed in Seoul's 'resolution for transport crisis' report
- Apr. 1992: Seoul directed by President Roh Tae-woo to build Phase 3 metro
- Apr. 1993: Seoul's Phase 3 metro development guideline reported to President Kim Young-sam and commissioned to the KRIHS
- Nov. 1993: Routes finalized: Line 9 – 12 planned, Line 3 extension
 - Routes extended into residential areas with poor access to urban metro, routes planned around 3 nuclei
 - Line 9 – 11: Out of 3 nuclei, two are mutually connected (Line 9 -Yeongdeungpo – Yeongdong; Line 10 -Yeongdeungpo-Yeouido – CBD; Line 11 -Yeongdeungpo – CBD)

- 1997: Reconsidered due to the Asian financial crisis
- Mar. 1998: Line 9 & extended Line 3 converted into metropolitan rail; all other routes canceled
- 2008: Some parts of Line 10 – 12 replaced with light rail
- 2009: Line 9 opened for service

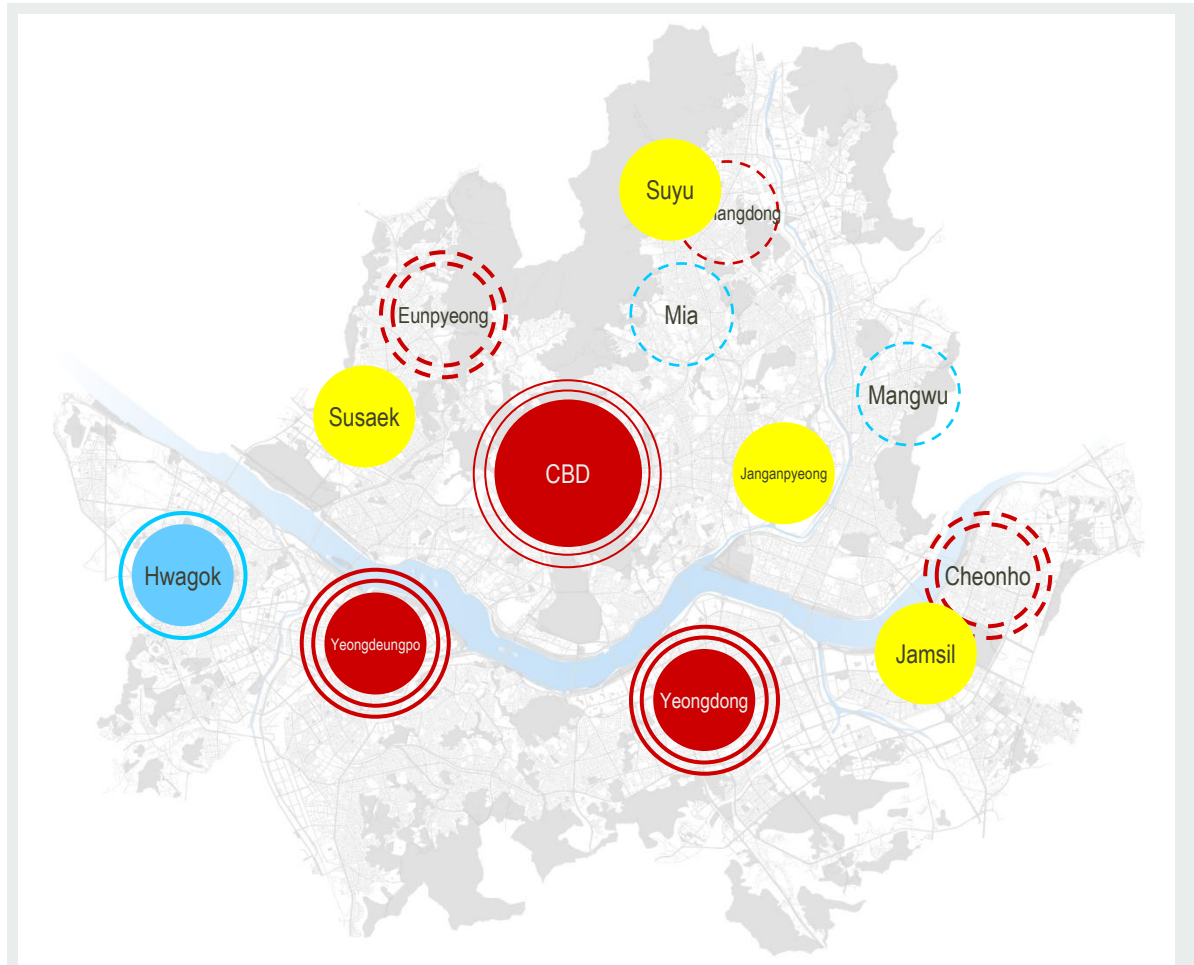
1960s



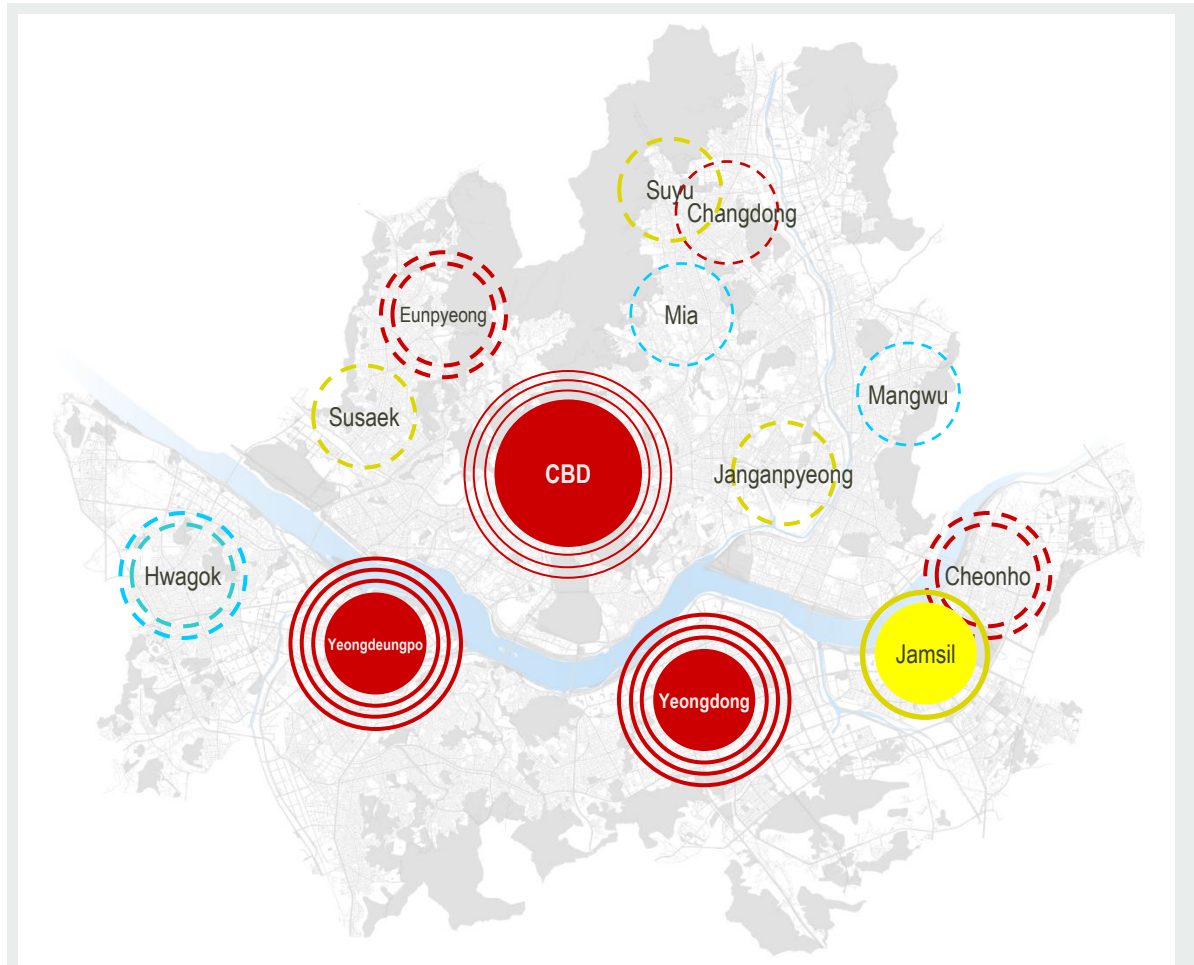
Early 1970s



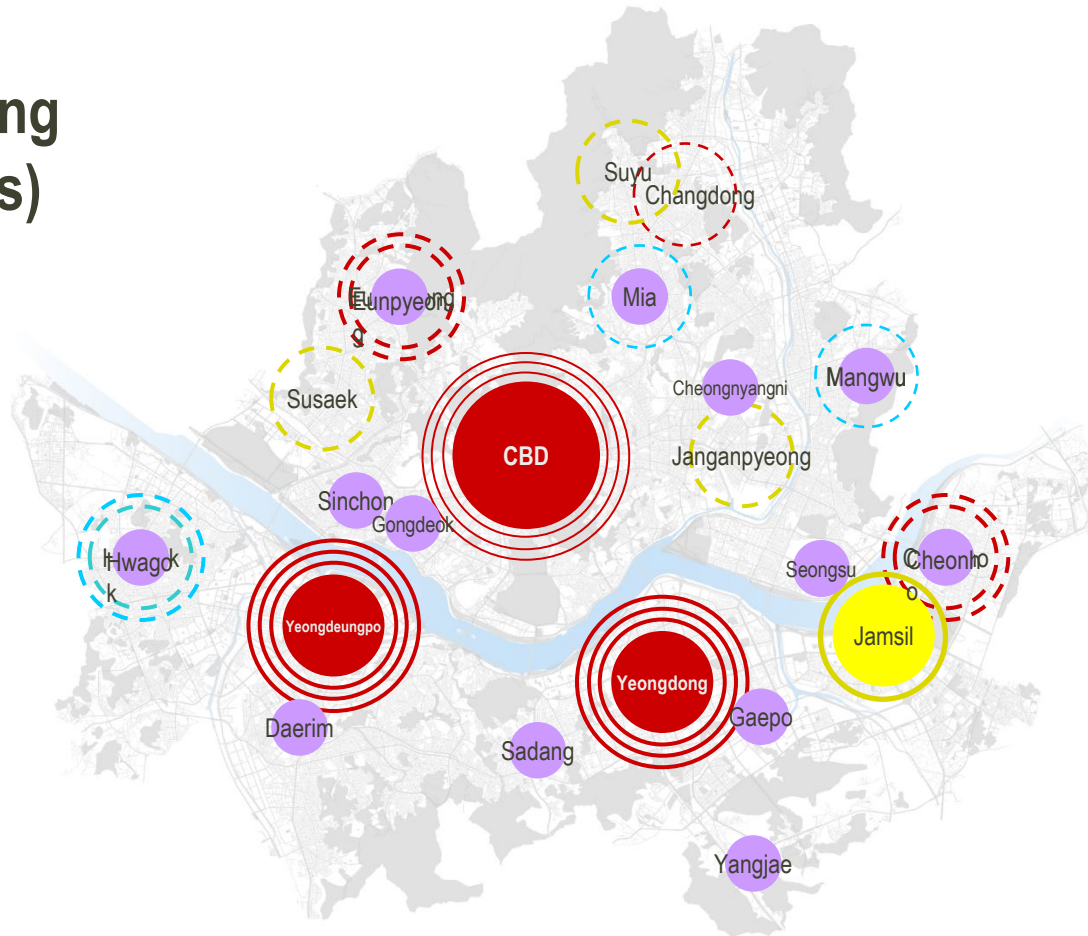
1978, 1980



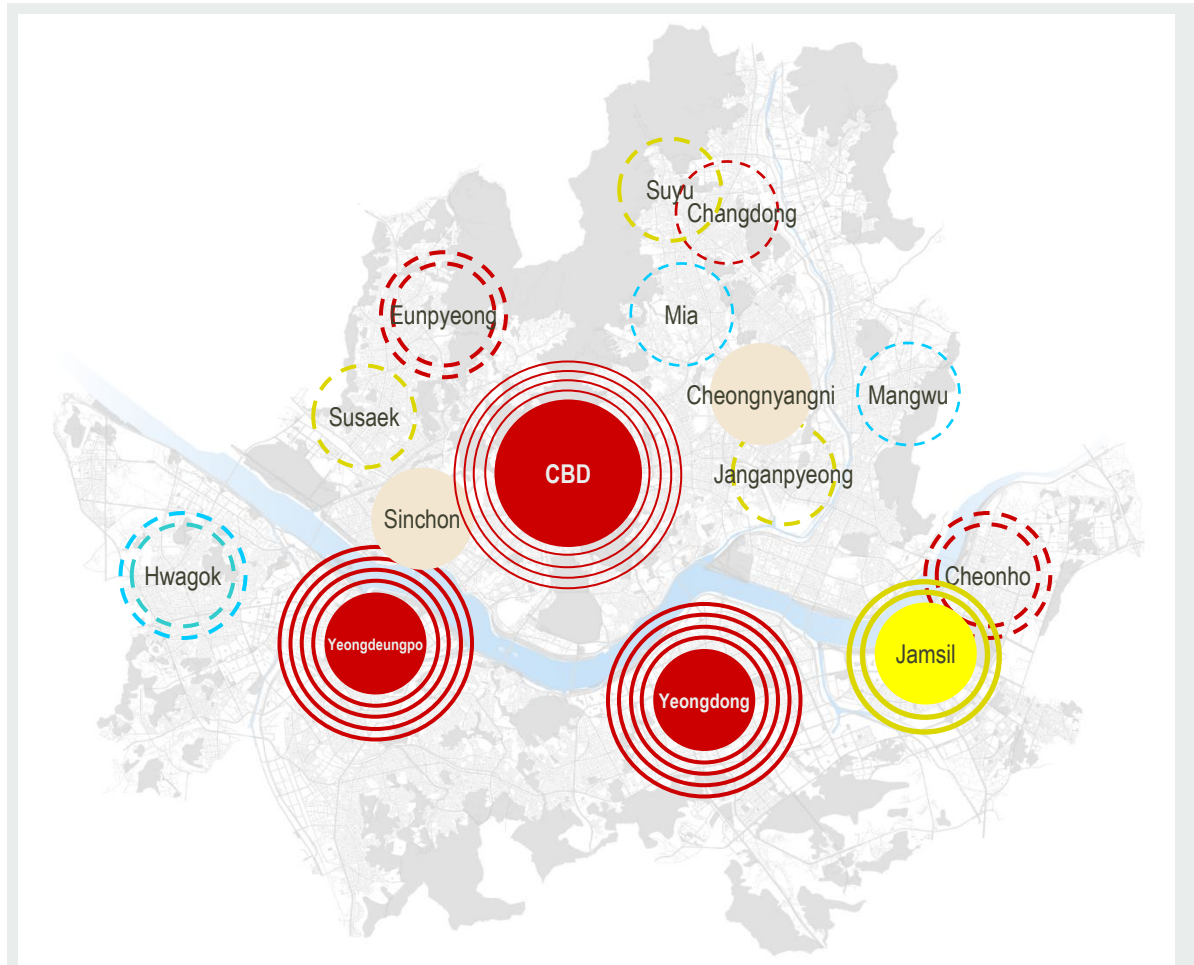
1984



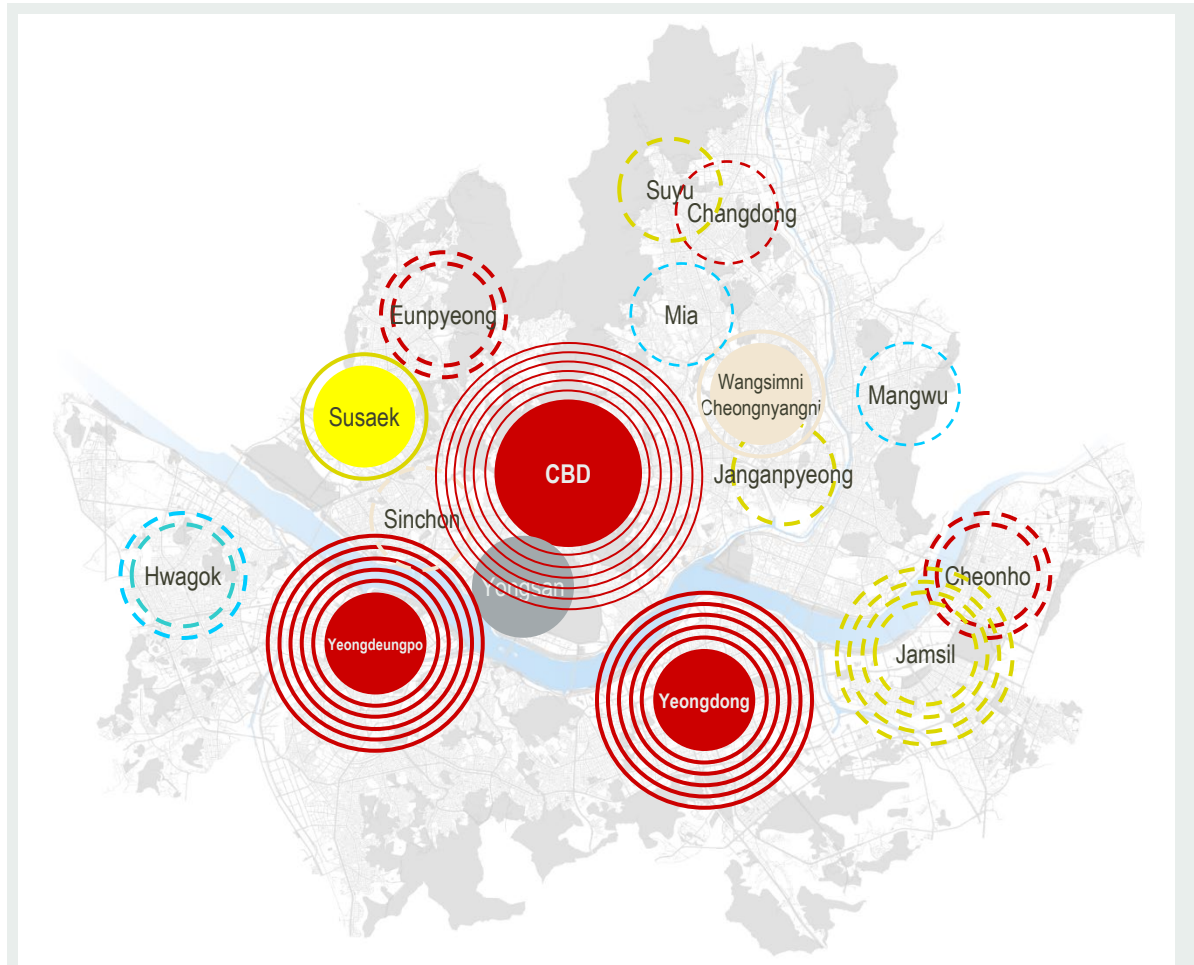
1984 (including
minor centers)



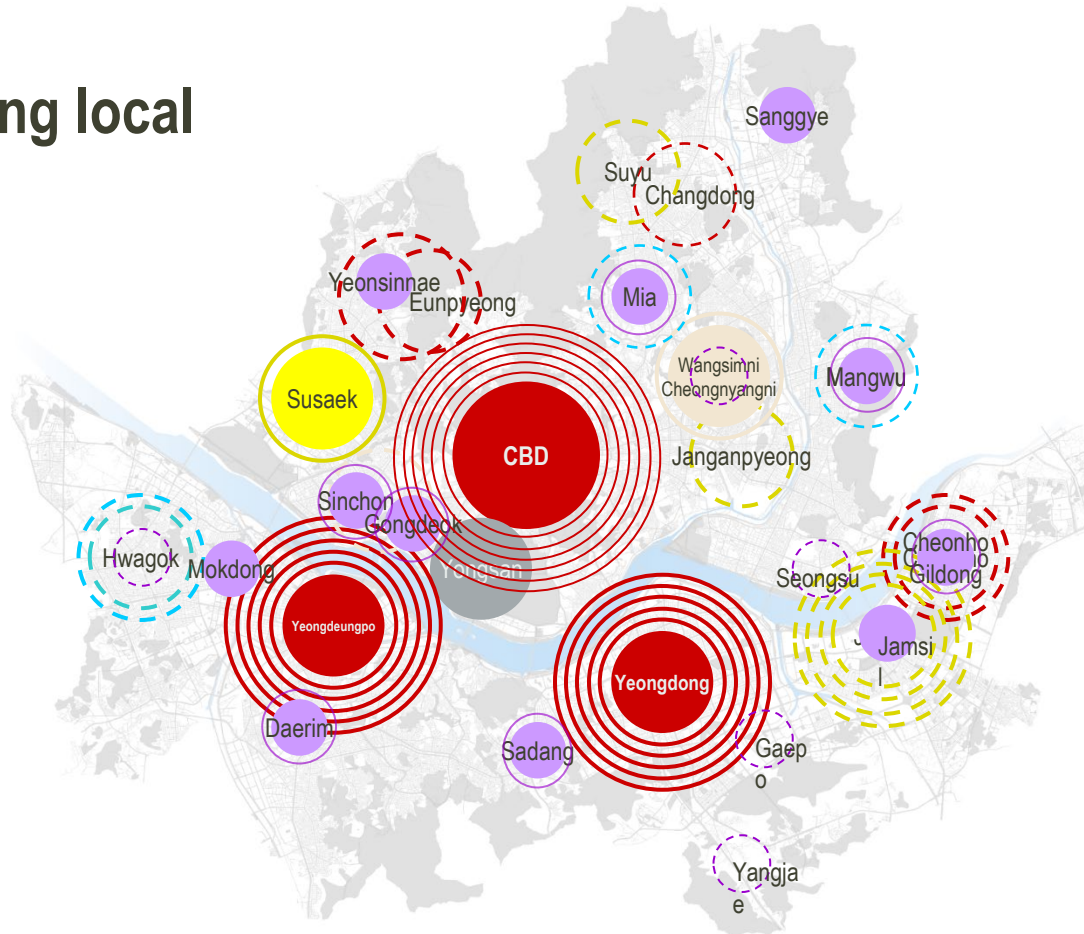
1990



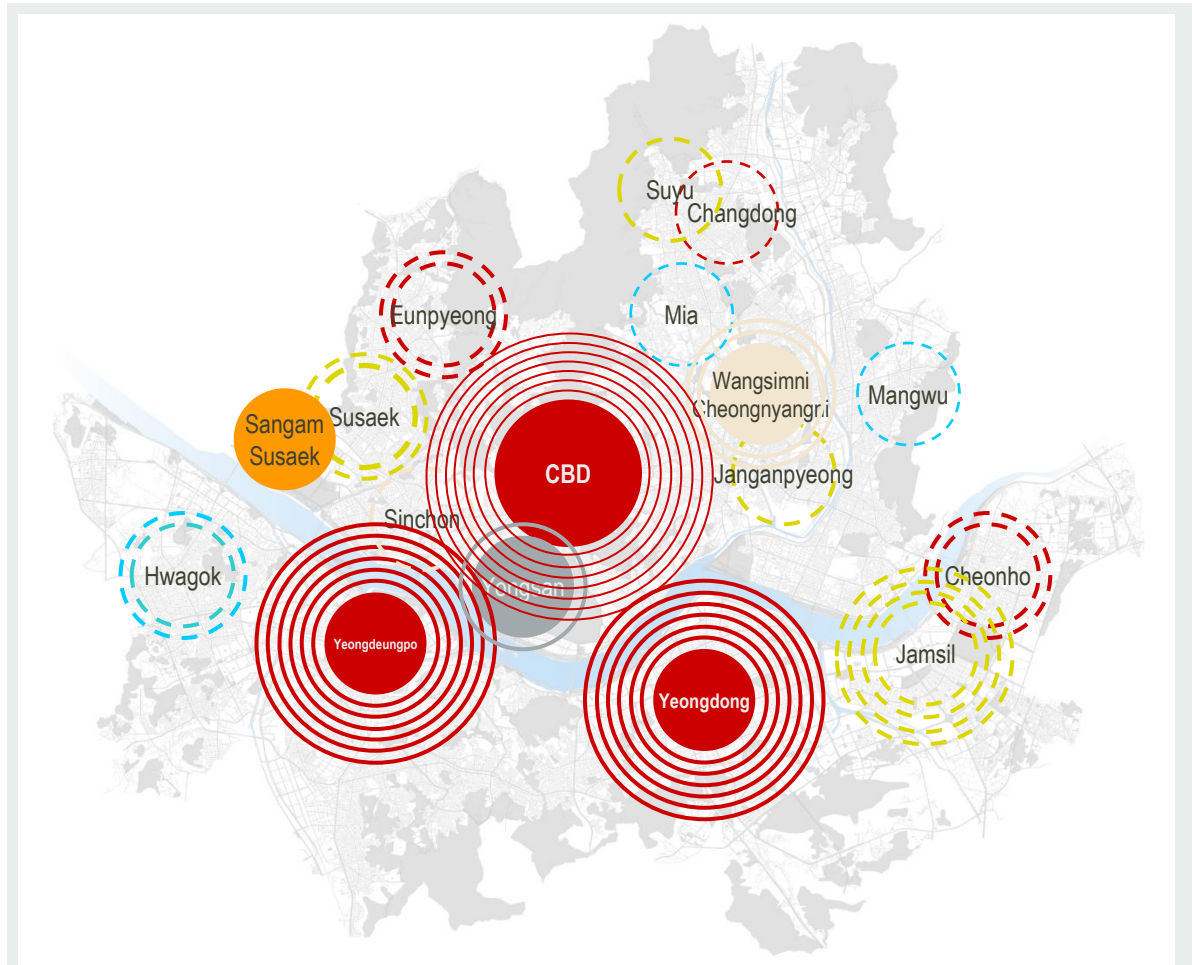
1997



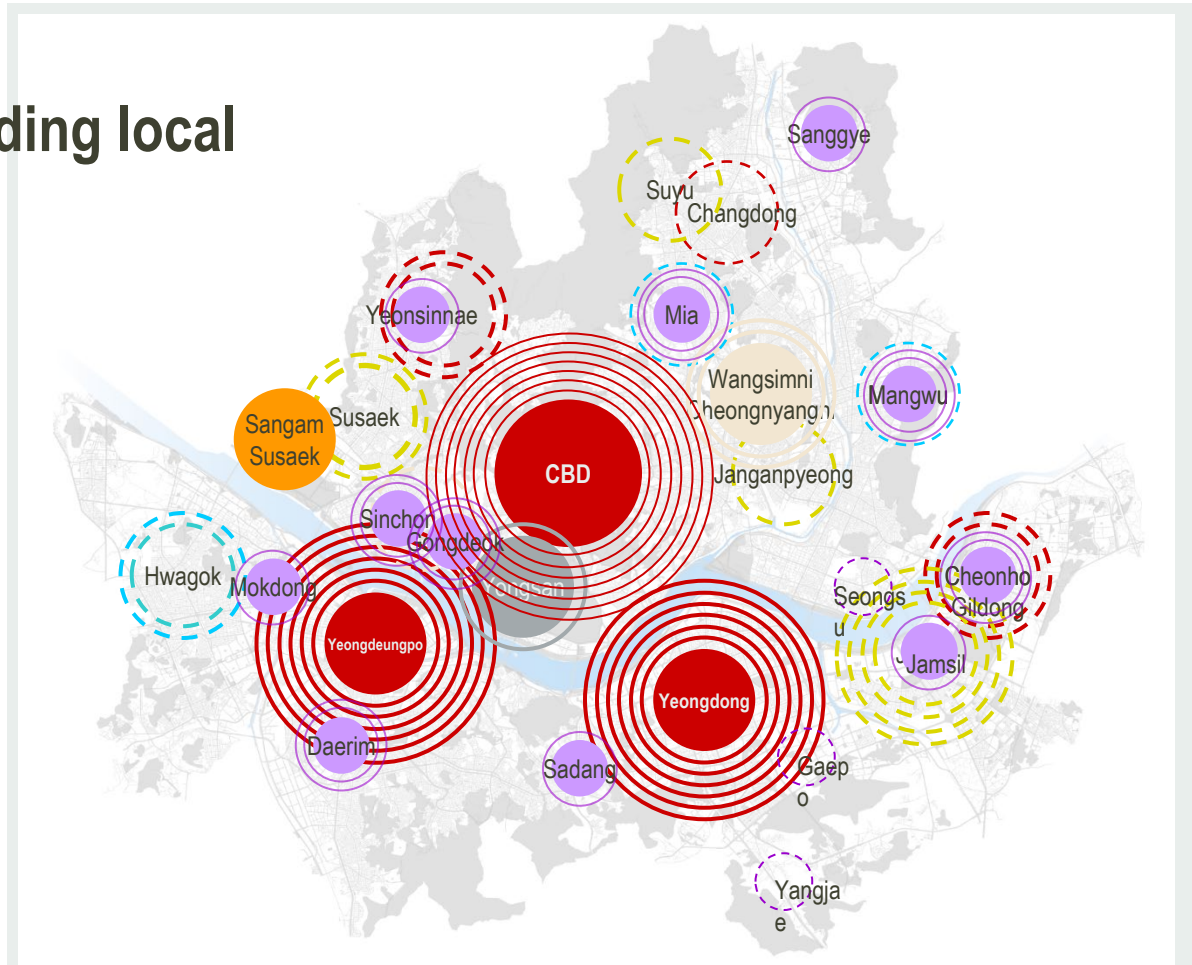
1997 (including local centers)



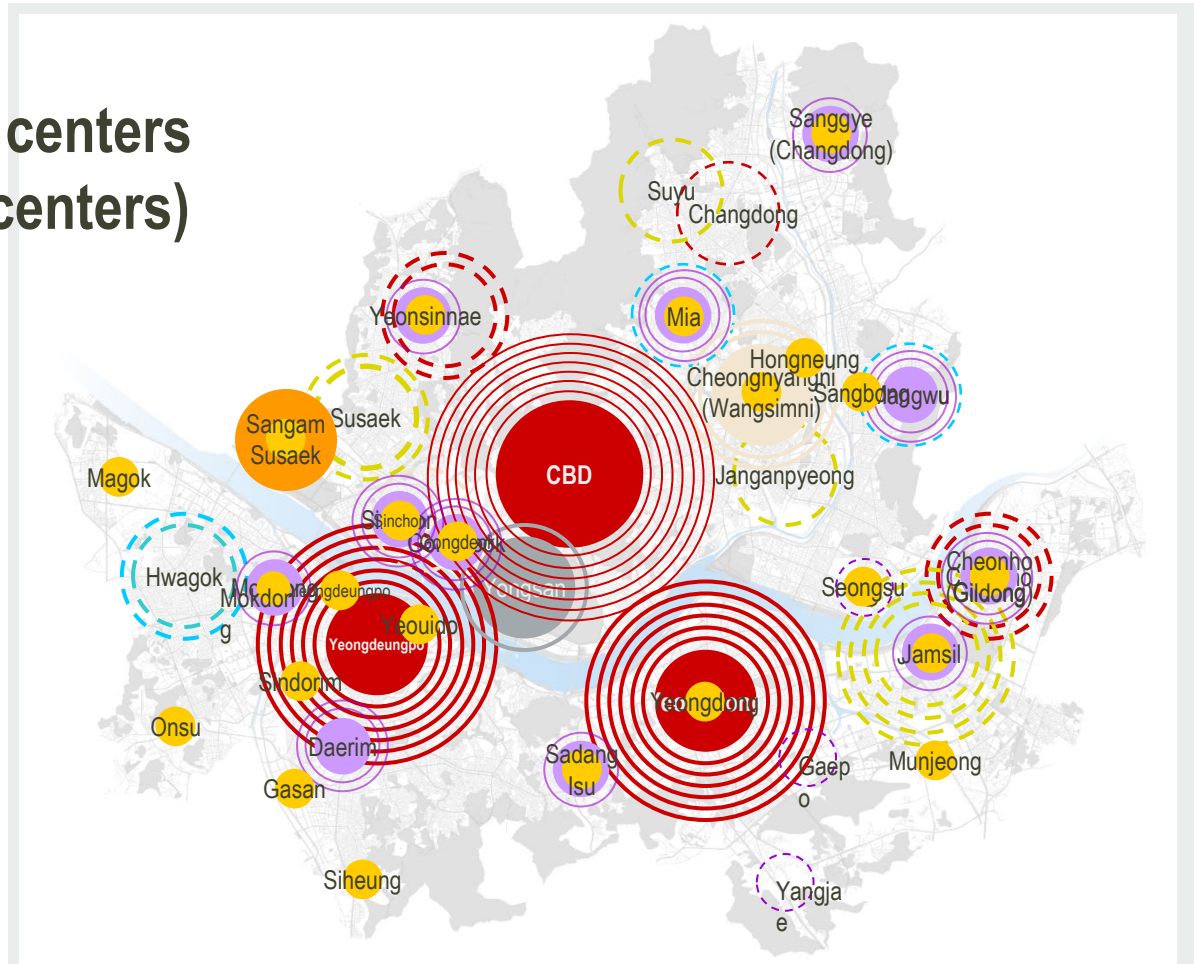
2006

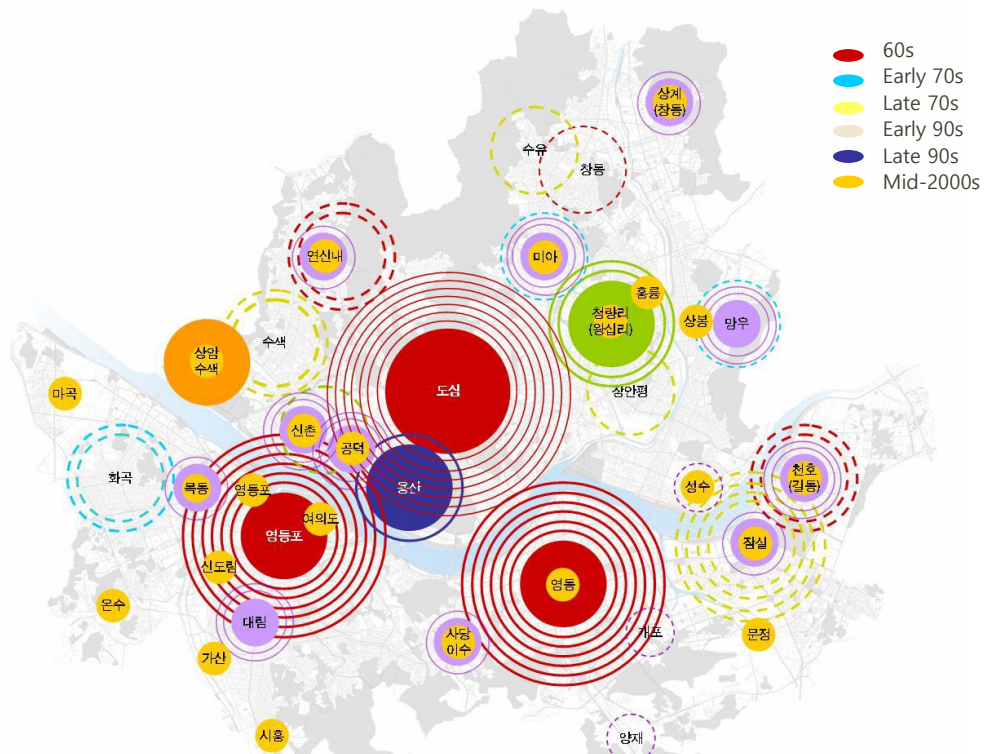
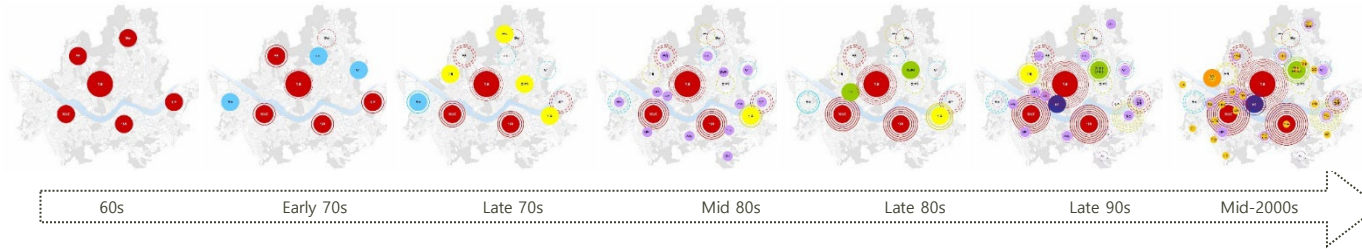


2006 (including local centers)



2006 (local centers + regional centers)





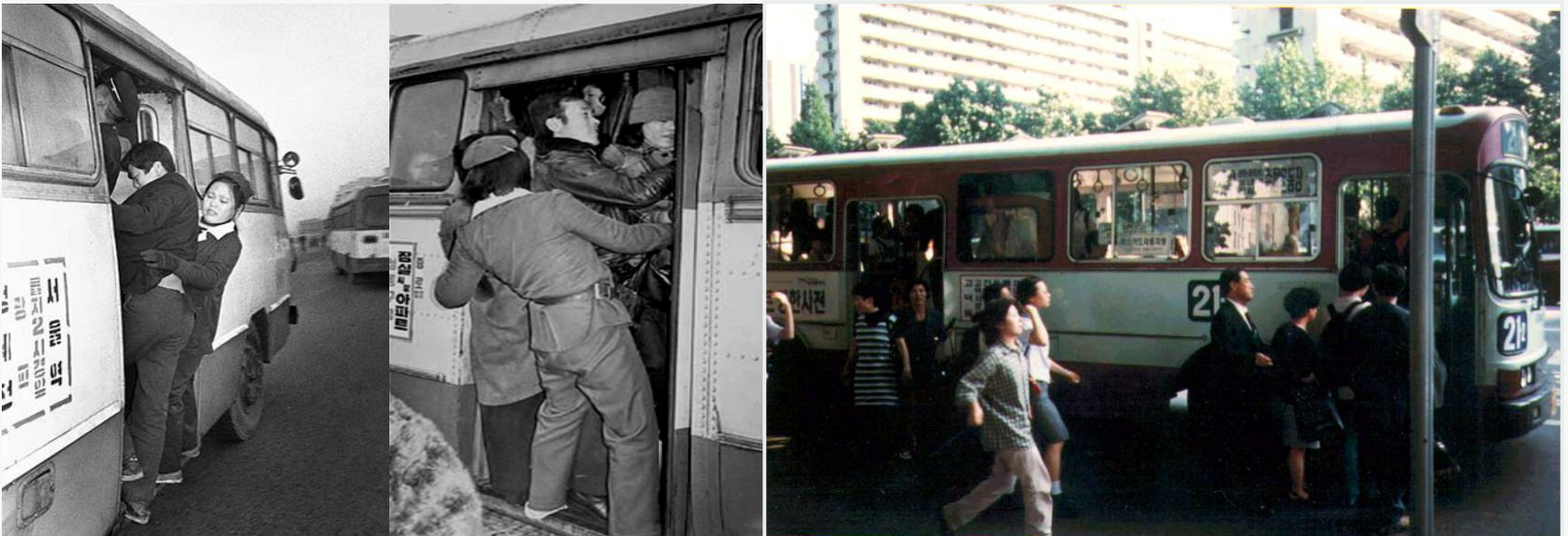
Planned / changed into a 3-CBD structure: CBD, Yeongdeungpo, and Gangnam

Internal data, The Seoul Institute (2018)

The Decline of Bus & the Rise of Subway

Bus Hit Hard by Subway System

- ✓ In the late 20th century, buses which practically lost competition with subways finally became reduced to a supplementary role for the subway system
- ✓ The 1997 Asian financial crisis also made citizens use public transit rather than driving their own cars, most of whom preferred subways to buses
- ✓ Bus had an important role to fulfill transport demand gap that subway cannot handle



Revival of Bus: Seoul Bus Reform in 2004

Initial Railway Plan

1st : Line 1 – 4

2nd : Line 5 – 8

3rd : Line 9 – 12 → **Unrealized due to 1997 IMF Crisis**

In 2002, “Task Force for the Activation of Mass Transportation” to search for effective solutions to improve bus system.

The “Citizens’ Committee for Bus Reform” was formed to garner the opinions of various interest groups.

The entire bus service network served the public good in an efficient and fair manner while still maintaining the bus companies as private entities.

- ✓ **The Quasi-Public Bus Operation System**, Integrated Fare System, **Integrated Smart Card System**
- ✓ Bus Speed guaranteed by the **Median Bus Only Lanes**
- ✓ State-of-the-Art Information System: **TOPIS**
- ✓ Bus System **Regionally** Integrated, Implementing New Bus System with Citizen Participation

BRT as an alternative



RED BUS 빨강(광역)버스



서울과 수도권 도시를 균형으로 연결하는 광역버스입니다.

BLUE BUS 파랑(간선)버스



서울 시내 먼거리를 운행하는 간선버스입니다.

GREEN BUS 초록(지선)버스



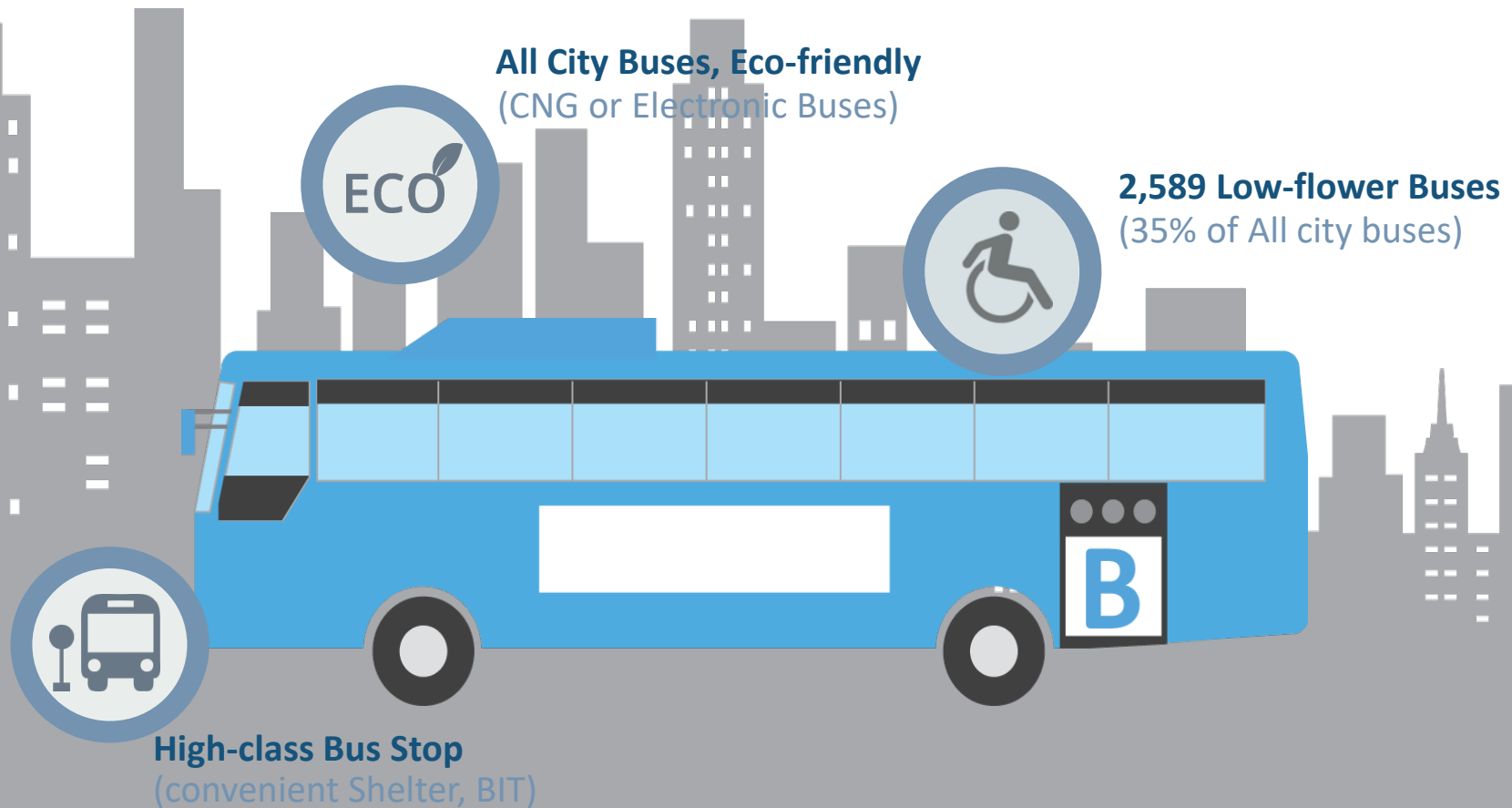
간선버스 · 시하철로 연계 및 지역내 통행을 위한 지선버스입니다.

YELLOW BUS 노랑(순환)버스



다양한 동행수요 대응형 순환버스입니다.

Bus System in Seoul



City Bus

390 routes, 7,855 buses
6,064 bus stops
19,910 bus drivers



5.8mil. /day



Night Bus

00-05

8 routes, 47 buses



Safe Return

-Choice of Place to get off
near home after 11 PM
- Return home with Safe Scout



Reorganizing bus routes and numbering system

(Sub)Urban areas ↔ Center cities

Meet the demand of passenger car

Connecting suburban areas and center cities

Meet the demand of passenger car

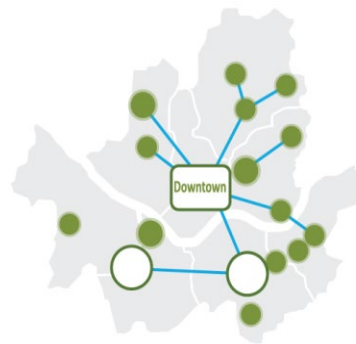
Link trunk line buses or subways for easy transfer

Satisfy the local needs and secure accessibility

Circular bus service for business
in urban areas



Inter-regional Lines



Trunk Lines



Feeder Lines



Circular Lines



Median Only Bus Lanes



For faster, reliable & Punctual bus



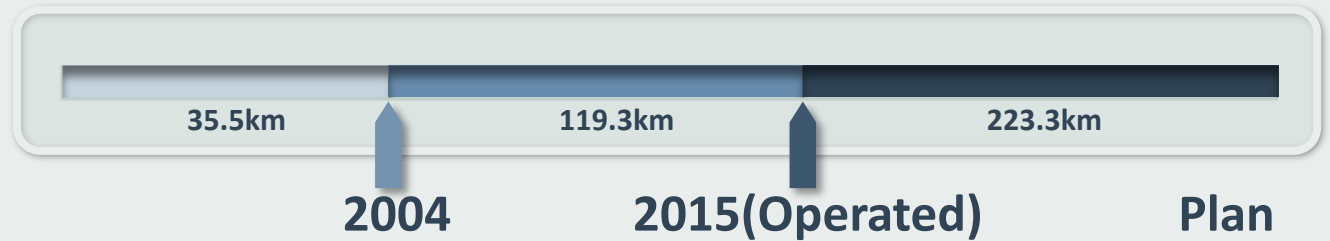
Median Only Bus Lanes



For faster, reliable & f



**12Corridor,
119.3km**



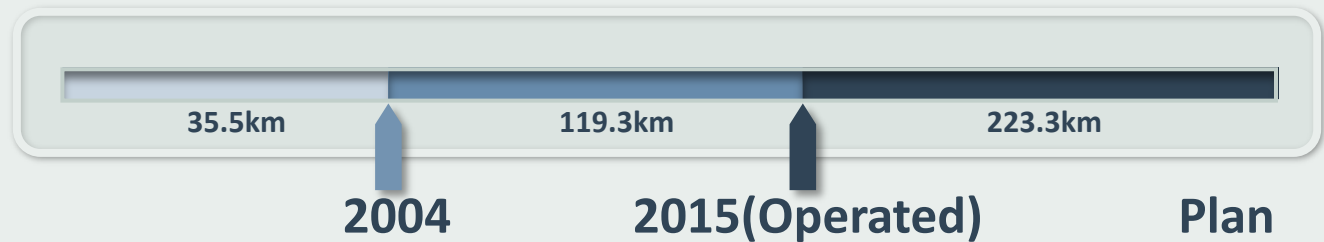
Median Only Bus Lanes



For faster, reliable & Punctual bus



12Corridor,
119.3km



Bus Speed

33%



2004

15km/h



2014

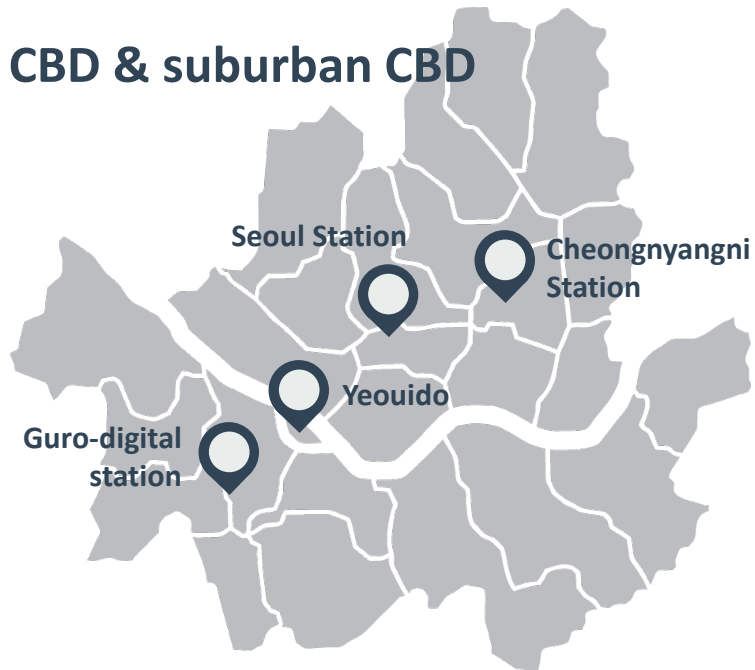
19km/h

Variance in
Operation Time

± 1-2min

Public Transit Transfer Center

In CBD & suburban CBD

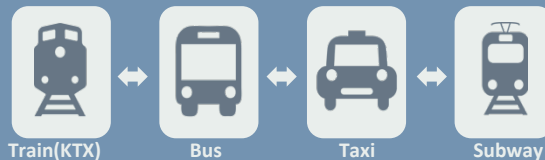


Seoul Transfer Center

 **80,000** person/day

Number of passengers for transfer

Modal, able to transfer



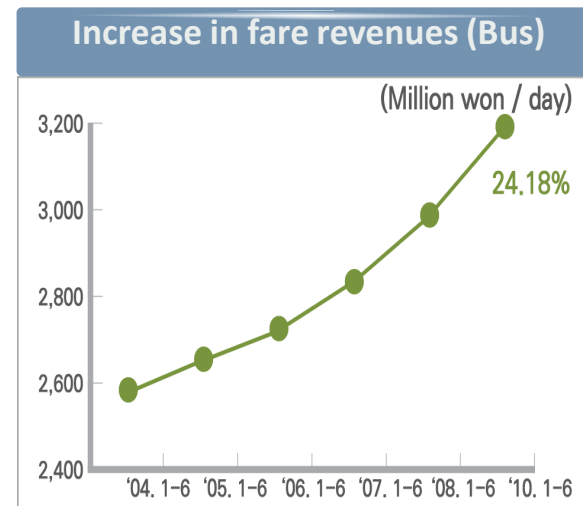
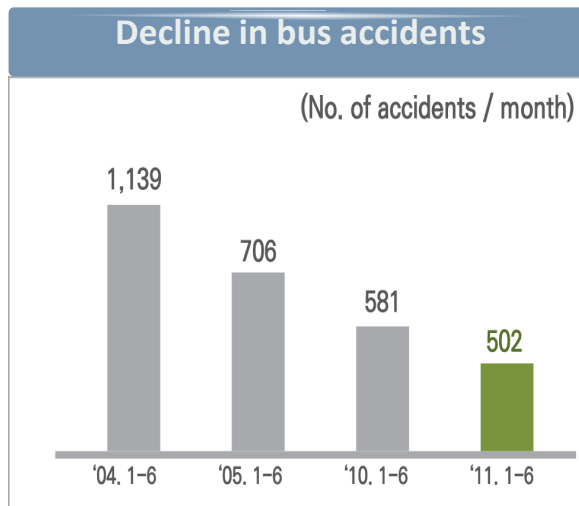
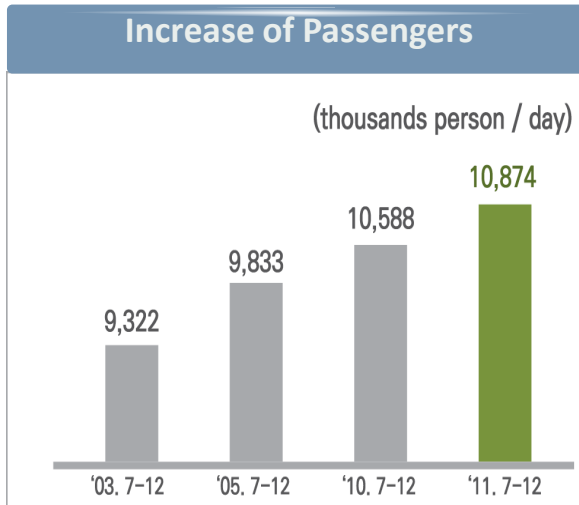
Time for Transfer

12min  **3min**



Major Achievement in Statistics

Social benefits expected : \$ 1.4 billion



Smart Card(T-Money) & Transfer System

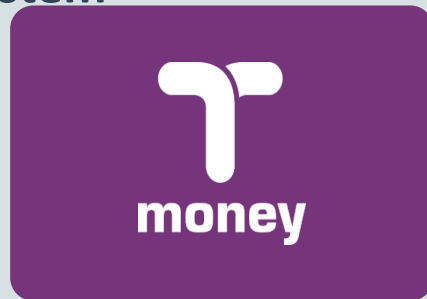


Free transfer bus-bus, bus-subway in Capital Region,

free transfer within 30 minutes (maximum of 5 times)



New Fare System



One Card, All Pass

Train(KTX)



Express Bus



Express way



Any Modal



Taxi

City Bus



Subway



NFC + Mobile



Anywhere



Integrated distance based fare system



5km, 1500won



4km, 0won

Total 1,500 Won (Basic Rate Within 10km)



10km, 1,550won



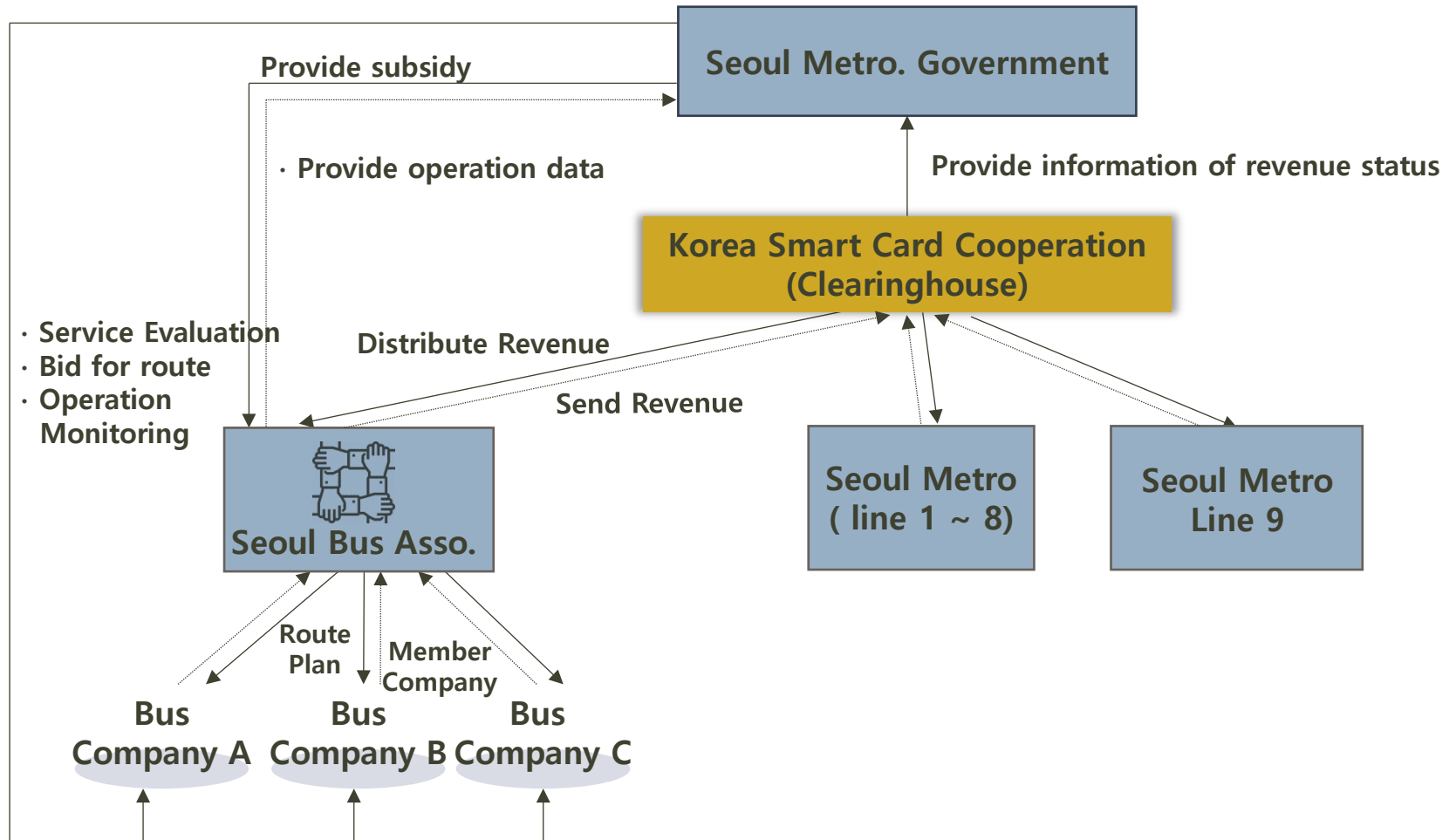
4km, 100won

Total 1,550 Won
(Basic Rate 1,550won of Subway +
Additional Rate 100won for 10km to 15km)



Card Usage Rate('15)

Transparent Management of Fare Revenues



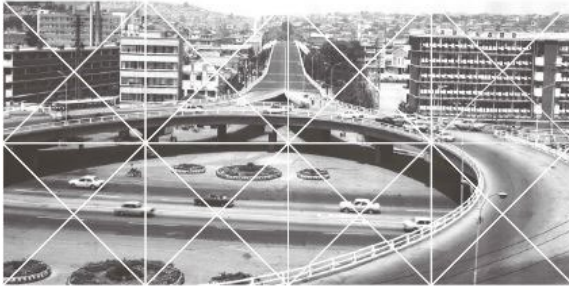
2017-09-21

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서울연구원
The Seoul Institute

2018-09-02

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